



Red teaming large language models: A comprehensive review and critical analysis

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ABSTRACT

Securing large language models (LLMs) remains a critical challenge as their adoption across various sectors rapidly grows. While advancements in LLM development have enhanced their capabilities, inherent vulnerabilities continue to pose significant risks, exposing these models to various forms of attack. This study provides a comprehensive review of LLMs' red teaming, distinguished by its broad coverage and intuitive organization. It systematically explores a range of red teaming attacks, including prompt-based attacks, data manipulation attacks, model exploitation attacks, information extraction attacks, and model degradation attacks. Additionally, it provides a critical review and analysis of evaluation methods and benchmarks, focusing on datasets, evaluation metrics, and benchmarking techniques used in LLM red teaming and risk assessment. Our review reflects the current state of LLM security and provides new insights alongside established methods by integrating recent and impactful research. The structured presentation of our findings offers a comprehensive and actionable resource, facilitating a deeper understanding of the complexities involved. This review highlights the proactive assessment of risk and exploitation potential, and contributes to the development of more secure and responsible LLMs, serving as a valuable guide for researchers, practitioners, and policymakers.

1. Introduction

The growth integration of large language models (LLMs) across diverse domains, ranging from healthcare to finance, transportation and entertainment, highlights the critical need for these models to operate safely and responsibly (Huang, Sun et al., 2024; Ozkaya, 2023). Ensuring the ethical and secure deployment of LLMs is essential to prevent potential harms (Hadi et al., 2023), such as the spread of misinformation (Sun et al., 2024), biased decision-making (Echterhoff et al., 2024; Eigler & Händler, 2024), or privacy violations (Yao et al., 2024). The remarkable capabilities of LLMs can be attributed to several factors, including the use of large-scale raw text datasets and advanced training architectures. For instance, models like GPT-4 (Achiam et al., 2023), with an estimated 1 trillion parameters, and PaLM (Chowdhery et al., 2023), trained on over 700 billion tokens, demonstrate the power of extensive data and complex architectures. However, as these models become more integrated into real-world systems, they

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Fig. 1. (a) The role of the red, blue, and purple teams in LLMs security. (b) The process of red teaming in overall workflow for securing LLMs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

also expose new and significant risks. Exploitation to adversarial attacks, privacy breaches, biases in training data, and unintended harmful outputs pose substantial challenges, making the security and safety of LLMs a critical area of concern.

To address these risks, red teaming has become an indispensable practice in evaluating and strengthening the security of LLMs. Traditionally used in cybersecurity, red teaming for LLMs involves simulating adversarial attacks and misuse scenarios to identify vulnerabilities in LLMs (Perez et al., 2022). By rigorously testing these models, red teams can uncover weaknesses before they are exploited in real-world situations (Ma et al., 2024). In this offensive role, the red team rigorously tests the model, seeking to identify weaknesses and inform improvements. It is essential as the landscape of attack techniques continues to evolve, with adversaries devising increasingly sophisticated methods to compromise LLMs. This proactive approach not only helps enhance the security of LLMs but also fosters trust and accountability. Complementing the red teaming, blue teaming focuses on defensive strategies as it is responsible for detecting, mitigating, and preventing attacks, thereby strengthening the system's overall resilience. On the other hand, purple teaming represents a collaborative integration of these approaches by compounding the insights of both red and blue teams to create a more holistic and adaptive security strategy. In Fig. 1(a), the red circle represents the proactive attack simulations of the red team, the blue circle represents the defensive measures of the blue team, and the overlapping purple area represents the integrated efforts of purple teaming in the context of LLMs. Whereas, the detailed workflow of the red teaming process in securing LLMs is illustrated in Fig. 1(b).

The primary motivation behind this review is to address the growing need for a comprehensive understanding of threat exploitation and attacks that LLMs face. While some existing reviews have touched upon specific vulnerabilities or attack vectors, there remains a lack of a systematic and unified perspective that encompasses the full spectrum of red teaming and benchmarks for LLMs' security.

Our motivation is further driven by the increasing reliance on LLMs in sensitive domains such as healthcare (Yang, Tan et al., 2023), finance (Li et al., 2023), and security (Motlagh et al., 2024), where the consequences of successful attacks can be severe. As LLMs are deployed in these high-stakes environments, it is crucial to ensure that they are not only effective but also safe and trustworthy. This review seeks to contribute to that goal by identifying key exploitation and attack strategies, with a comprehensive evaluation of their effectiveness, and providing a roadmap for future research and development in LLM red teaming for better security.

Additionally, the evolving landscape of LLM technology necessitates a continuous reassessment of risks and defenses. With the introduction of more advanced models, new attack vectors are likely to emerge, and existing defenses may become outdated. This review aims to stay ahead of these developments by synthesizing the latest research, offering insights into the current state of LLM security, and highlighting areas where further investigation is needed. Fig. 2 provides a comprehensive outline of the key aspects of LLM red teaming considered in this review. Ultimately, our motivation is to promote a deeper and complete understanding of the risks associated with LLMs and to support the development of more resilient and secure systems that can better withstand adversarial challenges.

The key contributions of this study are as follows:

- **Comprehensive exploration of LLM red teaming:** This study offers a detailed and systematic examination of the red teaming landscape for LLMs, covering a wide array of attack types and techniques, along with their strengths and challenges, surpassing existing reviews in both scope and depth.
- **Novel taxonomy and classification:** We introduce an intuitive taxonomy for categorizing LLM red teaming attacks, providing a more structured and accessible framework that clarifies the complex landscape of LLM security testing.

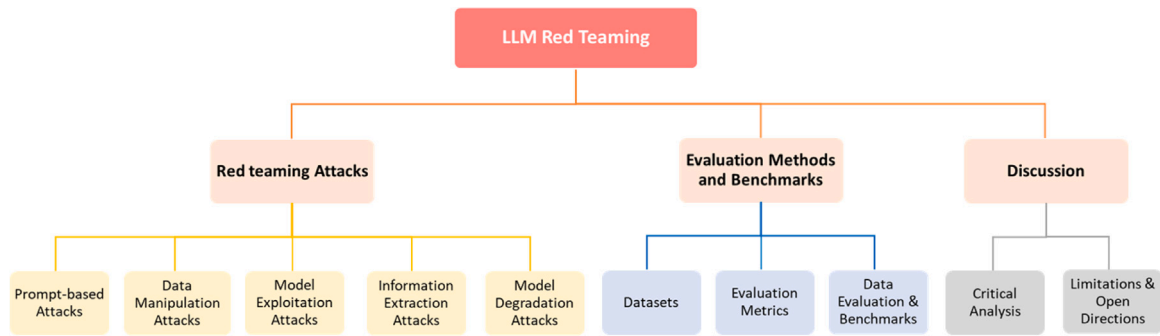


Fig. 2. The overview of LLM red teaming key aspects considered in this review. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

- *Advanced evaluation frameworks:* We analyze the evaluation frameworks, including datasets, metrics, and benchmarks used in LLM red teaming and its assessment. This systematic analysis enhances the effectiveness of security testing and promotes realistic threat simulation.
- *Updated state of the field:* This study integrates the latest developments in LLM red teaming, addressing the most pressing issues and providing a broader, more complete understanding of LLM security challenges, offering up-to-date information that surpasses existing reviews.

The rest of this article is organized as follows. Section 2 describes the review methodology, detailing the scope, comparison with related reviews, and the review process, including research evaluation criteria and questions. Section 3 provides the glossary of commonly used terminology in LLM security and comparison with related reviews. Section 4 examines various red teaming attacks on LLMs, categorized into prompt-based attacks, data manipulation attacks, model exploitation attacks, information extraction attacks, and model degradation attacks. Section 5 covers evaluation methods and benchmarks, assessing datasets, evaluation metrics, and benchmarking methods used in LLM red teaming and risk assessment. Section 6 presents a discussion of current research trends, along with a critical examination of the domain's limitations. Section 7 explores the broader implications of this study and its potential impact. Finally, Section 8 concludes the article.

2. Review method

Our review method is structured as follows:

- **Search Strategy Design:** We designed a search strategy that includes targeted search phrases and selected databases to identify relevant studies.
- **Study Selection:** We applied study selection criteria to filter the initial set of studies and selected the most relevant papers for in-depth analysis.
- **Quality Assessment:** We assessed the quality of the selected studies based on defined assessment criteria to ensure the credibility and reliability of the information.
- **Data Synthesis:** We extracted relevant data from the selected studies to address our research questions and provide a comprehensive analysis.

By following this structured review process, we aim to provide a thorough and systematic analysis of LLM red teaming, addressing the latest threats and evaluation frameworks, and offering valuable insights for researchers and practitioners.

2.1. Research questions

The primary goal of this study is to conduct a comprehensive and up-to-date review of red teaming in LLMs. To guide our analysis, we formulated the following research questions:

- **RQ1:** What is the current state of research in LLM red teaming?
- **RQ2:** What taxonomy could be used to represent and classify LLM red teaming attacks?
- **RQ3:** What are the major developments in the LLM teaming attacks?
- **RQ4:** How can effective LLM red teaming exercises be developed?
- **RQ5:** What are the key failure modes that lead to risks and vulnerabilities in LLMs?
- **RQ6:** What datasets and metrics are used for LLM red team attacks?
- **RQ7:** What evaluation frameworks and measures are employed to assess LLM red teaming effectiveness?
- **RQ8:** What are the major considerations for trustworthiness evaluation of LLMs, to ensure their reliable and ethical deployment in various applications?

Table 1

Checklist for our quality assessment criteria.

S. No.	Quality assessment questions
Q1	Does the paper have a well-defined methodology?
Q2	Are the LLM red teaming techniques sufficiently defined?
Q3	Are the performance measures fully defined and reported?
Q4	If applicable, is there a comparison with other approaches?
Q5	Does the study add/contribute to academia?
Q6	Based on the online publishing date, does the study have a sufficient number of average citations per year?
Q7	Is there information about the evaluation dataset that may be useful for gauging the success LLM red team attack?
Q8	Is there information about red teaming benchmarking methods that may be promising and scalable?

2.2. Search strategy

We conducted preliminary searches to identify relevant studies on LLM red teaming. Our search strategy involved the use of targeted keywords such as “Language Models” (OR “LLM”) AND “Vulnerabilities”, “Adversarial Attacks”, “Misuse”, “Ethical Risks”, “Data Poisoning”, and “Security and Privacy”. We searched several databases including Google Scholar, IEEE Xplore, ACM Digital Library, and arXiv. We further reviewed references and citations of key papers to uncover additional significant studies.

2.3. Study selection and quality assessment

The initial search yielded a large volume of papers, which were then filtered through a selection process. The papers obtained by our search were progressively shortlisted in order to identify the most relevant papers under the scope of our research questions. Initial screening involved reviewing titles and abstracts to determine relevance. Full-text reviews were conducted for studies that met the criteria to ensure they aligned with the focus of the review, as the criteria provided in Table 1. Studies were included based on their relevance to LLM red teaming, the risks posed, the impact of these risks, and the strategies for prevention and mitigation. We focused on studies that provide substantial insights into key aspects of red teaming LLMs; providing empirical results, theoretical insights, or methodological advancements. Considering the fact that the LLM red teaming is a fairly new domain, and many of the promising studies may still be under-review, we also considered the cited papers on ArXiv for our review.

2.4. Data synthesis

We defined a set of evaluation criteria to systematically assess and compare the reviewed studies:

1. *Red Teaming Type*: Identifies the specific red teaming approach used in the study (e.g., prompt-based attacks, data manipulation, model exploitation).
2. *Approach*: Describes the methodological approach, including theoretical framework, experimental setup, and whether the exercises are manual, automated, or hybrid.
3. *Modality*: Evaluates the modalities involved, primarily focusing on text-based LLMs and acknowledging cross-modal systems.
4. *Task*: Classifies the primary tasks addressed by red teaming efforts (e.g., adversarial attacks, misuse detection, harmful content detection).
5. *Techniques*: Examines the specific techniques employed, including data preparation, feature engineering, and modeling approaches.
6. *Dataset*: Assesses the datasets used, focusing on their sources and procedures.
7. *Generic vs. Specific*: Determines whether the methods are generalizable or tailored for specific applications.
8. *Language*: Evaluates the linguistic scope of the attacks, including studies exploiting vulnerabilities in low-resource languages.
9. *Evaluation Methods and Metrics*: Assesses the methods and metrics used to evaluate red teaming efforts.

This process forms the foundation for our comprehensive review, enabling a structured comparison and critical analysis of the studies.

2.5. Scope

Our review aims to provide a comprehensive analysis covering a wide range of red teaming aspects, including vulnerability exploitation, attack types, evaluation methods, and benchmarking frameworks. We have also organized these aspects into coherent taxonomies and organization for a better understanding of the field, and we focused on significant and novel findings while excluding studies with marginal contributions.

3. Background

This section begins by defining key terms frequently encountered in LLM red teaming. We then examine related reviews to identify their scope, strengths, and limitations, which motivate the comprehensive approach of our study.

Table 2
Terminology and descriptions for frequently used concepts in LLMs' safety and security.

Terminology	Description
Alignment	Alignment ensures that a system's behavior and outputs adhere to set values and ethical principles. This process often involves fine-tuning and reinforcement learning with human feedback (RLHF) and other methods to meet specific ethical standards and criteria.
Association	The ability of LLMs to form connections between different pieces of information, such as words, entities, concepts, or events. This can enable applications like knowledge retrieval and text summarization, but can also pose challenges such as privacy leakage and bias.
Auditing	The process of examining LLM memorization to understand its implications and consequences. For instance, auditing verbatim memorization involves generating arbitrary strings to detect if the LLM can reproduce them with high probability.
Bias	The presence of systematic and unfair discrimination in the model's predictions or behavior, often reflecting prejudiced assumptions in the training data. Bias in LLMs can lead to unfavorable or inaccurate outcomes for certain groups.
Black-box attack	An attack lacking direct access to the model's internal state and details. Instead, it interacts with the model through provided APIs and devise strategies based on the model's outputs.
Faithfulness	Refers to how accurately a model's output reflects its input or aligns with provided external evidence (e.g., documents in a RAG system). A response is deemed faithful if it accurately reproduces or summarizes the information contained in the prompt or retrieval context, without adding extraneous details from its own learned knowledge. For instance, if a retrieval-augmented LLM is given a passage on a historical event, a faithful summary would include only details present in that passage and avoid hallucinating additional facts.
Hallucination	Refers to instances when a model generates content not grounded in factual reality accessible within its knowledge, leading to incorrect information or fabricated facts. This often results from misinterpretations due to over-generalization, knowledge gaps from data sparsity, pattern recognition errors, or difficulty with complex or ambiguous queries. Hallucinations reflect a breakdown in faithfulness, as the model produces content unsupported by its knowledge, resulting in false details or invented data.
Harmful behavior	Refers to intended or unintended actions and outputs that expose users to offensive, unsafe, unreliable, or ethically concerning content. Such behaviors include producing toxic or offensive language, revealing sensitive or private information, perpetuating harmful biases, and generating untruthful or unfaithful content.
Memorization	The phenomenon of LLMs retaining and reproducing information from their training data. Memorization can be beneficial for tasks requiring factual knowledge but also problematic for privacy, security, and quality reasons.
Red teaming	Red teaming is a structured process used to simulate adversarial scenarios to identify vulnerabilities and test the robustness of systems or models. This practice involves deliberately probing and manipulating models to discover potential harmful behaviors and weaknesses. The goal is to improve system security and resilience by anticipating potential threats and developing mitigation strategies. In AI safety, red teaming can employ various methods such as adversarial attacks, jailbreaking, and prompt injection to rigorously test model defenses and alignment with human values.
Robustness	The ability of an LLM to maintain its performance and reliability when faced with adversarial attacks, noisy data, or unexpected inputs. Robustness is crucial for ensuring the dependability of LLMs in real-world applications.
Trustworthiness	Encompasses the aspects of reliability, transparency, safety, fairness, and ethical alignment in LLMs. A trustworthy model consistently produces correct, unbiased, and safe outputs with transparency in its decision-making process, while clearly mentioning uncertainty when applicable. For example, in a healthcare application, a trustworthy LLM should reliably provide accurate medical information without generating misleading or harmful advice.
Truthfulness	Refers to the degree to which model-generated content accurately reflects factual information and avoids fabrications or misleading content. Based on the LLMs nature and their massive training data with mixed-factual accuracy, they often generate plausible-sounding yet incorrect or hallucinated details.
Watermarking	Embedding hidden signals in text generated by an LLM to make it algorithmically identifiable as synthetic while being imperceptible to humans. This helps mitigate risks like disinformation or plagiarism by proving the text's authenticity.
White-box attack	An attack based on complete access to the model's internal structure, including its architecture, parameters, and sometimes training data. This extensive access allows attackers to craft precise and highly effective adversarial attacks targeting the model's specific weaknesses.

3.1. Terminologies

We provide comprehensive definitions of the most crucial terminology used throughout this paper in [Table 2](#). Given the diverse interpretations of these terms across different sources, it is essential to detail clear and encompassing definitions to avoid any ambiguities. These definitions should help in understanding the nuanced differences and similarities in how these terms are used in the context of red teaming LLMs, as a quick reference guide.

3.2. Related reviews

Despite the growing body of research, many existing reviews offer fragmented or narrowly focused insights into LLM security. Several studies concentrate on specific attack vectors or defense mechanisms without providing a holistic view of the interplay between different types of attacks and mitigation strategies ([Table 3](#)). This piecemeal approach often leaves gaps in understanding the broader implications of LLM vulnerabilities and their comprehensive mitigation. For instance, [Wu, Duan et al. \(2024\)](#) addresses application-based privacy threats specific to ChatGPT but lacks broader coverage of diverse attack types and defenses. [Gallegos et al.](#)

Table 3

Systematic comparison of our review with the related surveys on LLM security and red teaming. ✓: Covered, ⊖ : Partially covered, R: Review, T: Technical.

Authors	Highlights	Comprehensive	Systematic	Critical	Research type	General privacy issues	Jailbreaking attacks	Prompt injecting	Backdoor attack	Data poisoning attack	Gradient leakage attacks	Membership inference	PII Leakage	Other attacks	Evaluation	Benchmarks
Neel and Chang (2024)	LLM privacy threats	⊖		⊖	R	✓							✓		✓	
Huang, Sun et al. (2024)	Trustworthiness in LLM	⊖			R + T	✓								✓	⊖	⊖
Das et al. (2025)	Review of security attacks, privacy risks, and defenses	⊖		⊖	R	✓	⊖	⊖	⊖	⊖	⊖	⊖	⊖	⊖		
Anwar et al. (2024)	Foundational challenges regarding LLMs alignment and safety	✓			R	✓								⊖		
Abdali et al. (2024)	LLM privacy and security concerns	⊖		⊖	R	✓	⊖	⊖	⊖	⊖	⊖		⊖	⊖		
Cui, Wang et al. (2024)	Risk taxonomy, assessment, and mitigation	✓			R	⊖	⊖	⊖	⊖	⊖		⊖	⊖	⊖	⊖	⊖
Chowdhury et al. (2024)	Attacks on LLMs	⊖			R	⊖	⊖	⊖	⊖				⊖	⊖		
Lin et al. (2025)	Red Teaming for Generative Models	⊖		✓	R	⊖	⊖	⊖	⊖				⊖	⊖	⊖	
Wu, Duan et al. (2024)	Application-based privacy threats in ChatGPT models		⊖		R	✓	⊖	⊖		⊖						
Gallegos et al. (2024)	Bias and fairness in LLMs	⊖	⊖	⊖	R	✓									⊖	
Gupta et al. (2023)	ChatGPT Generative AI impacts on cybersecurity and privacy. 2023 topical review			⊖	R	✓	✓							✓		
Liu, Deng, Xu et al. (2024)	Overview of various jailbreaking attacks on ChatGPT			⊖	R		✓									
Yang, Xiang et al. (2024)	Various backdoor attacks in LLMs within communication networks				R				✓							
Lukas et al. (2023)	LLM PII leakage analysis	⊖	⊖	⊖	R + T								✓		⊖	
Yao et al. (2024)	LLM privacy and security	✓	⊖		R	⊖	⊖	⊖	⊖	⊖		⊖		⊖		
Dong, Zhou et al. (2024)	LLM conversation safety		✓		R	⊖	⊖	⊖	⊖	⊖				⊖	⊖	
Park et al. (2024)	Empirical study of AI deception		⊖	✓	R	✓										
Perez et al. (2022)	Exploration of language model based LLM red teaming		✓	⊖	R + T	⊖	⊖	⊖					⊖	⊖		
Hassanin and Moustafa (2024)	LLMs for cyber-defense	⊖	⊖	⊖	R									✓		
This review	Comprehensive review of LLM red teaming attacks and their evaluation	✓	✓	✓	R	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

(2024) provides insights into bias and fairness but does not address the full range of red teaming attacks or defenses. Dong, Zhou et al. (2024) and Gupta et al. (2023) offer topical reviews on ChatGPT impacts and conversation safety, respectively, but do not cover all relevant attack vectors or detailed risk assessments. Similarly, Hassanin and Moustafa (2024) and Perez et al. (2022) focus on red teaming and cyber-defense but may not fully address the spectrum of LLM vulnerabilities and mitigation strategies. Wang, Shibghatullah et al. (2024) highlights the importance of detecting cyberbullying but lacks comprehensive coverage of LLM security

issues. Additionally, [Lukas et al. \(2023\)](#) and [Yao et al. \(2024\)](#) contribute to privacy and PII leakage studies but lack a systematic approach to evaluating defenses and risk assessment methodologies. Furthermore, [Lin et al. \(2025\)](#) provides a survey on red teaming for generative models while highlighting critical vulnerabilities, but it lacks thorough and broader landscape coverage of LLM-specific attack vectors and techniques. Moreover, [Neel and Chang \(2024\)](#) discusses privacy threats but does not systematically cover various attack types or defenses. [Huang, Sun et al. \(2024\)](#) explores trustworthiness but only partially addresses the broad spectrum of security issues. [Das et al. \(2025\)](#) reviews security attacks, privacy risks, and defenses but lacks comprehensive coverage. [Anwar et al. \(2024\)](#) touches on foundational challenges in LLM alignment and safety without an exhaustive examination of attack vectors. [Abdali et al. \(2024\)](#) covers privacy and security concerns but only partially addresses various attacks and defenses. [Cui, Wang et al. \(2024\)](#) presents a risk taxonomy and assessment but does not fully cover all attack types in detail. [Chowdhury et al. \(2024\)](#) focuses on LLM attacks but lacks a systematic and comprehensive approach. [Liu, Deng, Xu et al. \(2024\)](#) overviews jailbreaking attacks but does not address other critical attack types. [Yang, Xiang et al. \(2024\)](#) examines backdoor attacks but overlooks other significant vulnerabilities. Finally, [Park et al. \(2024\)](#) provides an empirical study of AI deception but does not cover the complete range of LLM security issues. As most related reviews focus primarily on specific attacks or issues, with varying depths of coverage. This gap motivates us to undertake a thorough review of attacks, risk assessment, and benchmarking pertaining to LLM red teaming.

Some related reviews have proposed various taxonomies and frameworks for categorizing LLM red teaming attacks, each offering unique perspectives and classifications. However, many of these reviews lack comprehensive coverage for a fair comparative analysis of taxonomies. For example, some approaches are based primarily on developmental and deployment stages (such as data collection, model training, and deployment), while others focus on module-oriented risks (dividing LLM systems into components like input, language model, toolchain, and output), or adopt capability-based and socio-technical frameworks that consider the interplay between technical vulnerabilities and the social contexts in which LLMs operate. For instance, [Cui, Wang et al. \(2024\)](#) introduced a module-oriented risk taxonomy that dissects LLM systems into four essential modules, each associated with specific risks such as harmful content generation, untruthful outputs, and unhelpful uses. However, many red teaming attacks and their corresponding defenses often span multiple modules, which limits the practical application of such narrowly defined frameworks. In contrast, [Lin et al. \(2025\)](#) proposed a taxonomy grounded in the inherent capabilities of language models, categorizing attacks based on the specific functionalities they exploit. Collectively, these diverse taxonomies offer complementary frameworks for understanding LLM red teaming attacks, yet they tend to focus either on developmental stages, specific attack techniques, or literature surveys without providing a multidimensional perspective. A comprehensive taxonomy should present a structured overview of adversarial strategies that evolves into a holistic framework, one that not only categorizes a broad spectrum of attack vectors, but also informs the development of effective security measures for real-world applications. Our proposed framework presented in Section 4 emphasizes the multifaceted nature of threats; ranging from prompt manipulation to the exploitation of data and architectural vulnerabilities, and aims to bridge the gaps identified in existing taxonomies and frameworks.

As LLMs become increasingly integrated into real-world applications, new and sophisticated attack techniques continue to emerge. Existing reviews may not adequately cover recent advancements in attack methodologies or evolving defense mechanisms. Our review aims to bridge this gap by providing an up-to-date and thorough analysis of both established and emerging red teaming practices, ensuring that the latest threats and countermeasures are addressed.

Another critical aspect often overlooked is the integration of risk assessment and benchmarking in the context of LLMs. Effective risk management requires a nuanced understanding of various vulnerabilities and a structured approach to evaluating the effectiveness of mitigation strategies. By incorporating comprehensive risk assessment and benchmarking practices, our review provides a more complete framework for evaluating LLM security.

Our review stands out by offering an extensive and systematic evaluation of LLM red teaming studies. We cover a broad spectrum of attack types, from prompt-based attacks to more sophisticated methods, and cover risk assessment and evaluation methodologies. By integrating these elements, our review bridges gaps left by previous studies and provides a more complete understanding of LLM vulnerabilities and security strategies. As detailed in [Table 3](#), this review offers a comprehensive and in-depth analysis that distinguishes it from existing reviews.

Our review encompasses a wide range of topics, including vulnerabilities, attack types, defenses, risk assessments, and evaluations associated with LLMs in the context of red teaming. We aim to organize these aspects into coherent taxonomies for a comprehensive understanding of the field. By addressing a broad spectrum of attack methodologies and evaluating various mitigation strategies and benchmarks, our work provides valuable insights for researchers and practitioners. This comprehensive approach enhances the understanding of LLM security challenges and contributes to the development of more effective mitigation strategies, ultimately advancing the field of LLM safety and robustness.

4. Red teaming attacks on LLMs

Red teaming attacks on LLMs represent a critical area of research aimed at rigorously testing and evaluating the robustness, safety, and security of these advanced models. These attacks are designed to simulate adversarial behavior and assess misuse potential, to uncover vulnerabilities that could be exploited in real-world scenarios. By systematically probing the models across various attack vectors, red teaming provides invaluable insights into potential failure points and informs the development of more secure and resilient systems. In this section, we categorize red teaming attacks into five classes: prompt-based, data manipulation, model exploitation, information extraction, and model degradation Attacks, each representing distinct approach(es) to testing the robustness and security of LLMs. The infographic covering the categorized LLM red teaming attack techniques is presented in [Fig. 3](#), and [Table 4](#) presents tasks and evaluations of prominent LLM red teaming techniques.

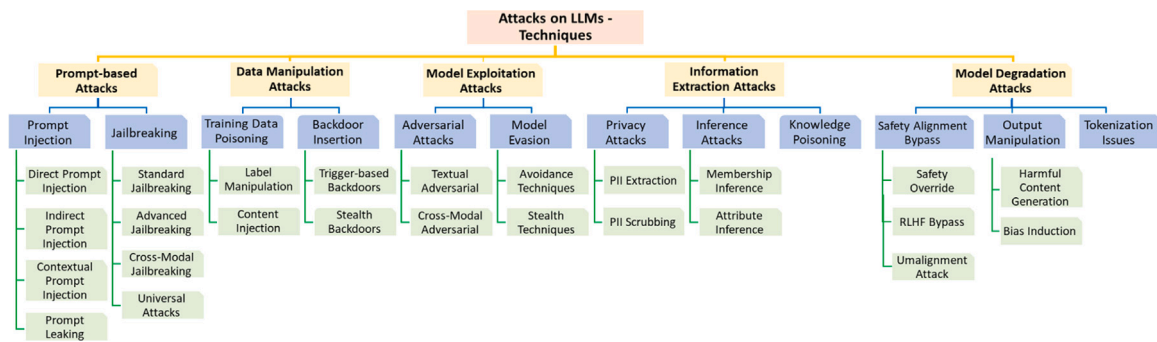


Fig. 3. Techniques and methodologies for red teaming attacks on LLMs. Shapes with rounded and/or snipped corners signifies training-time and inference-time attacks, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.1. Prompt-based attacks

Prompt-based attacks target the fundamental mechanism by which LLMs interpret and respond to user inputs. By manipulating the prompts given to an LLM, adversaries can alter the model's behavior leading to unintended sometimes harmful outcomes. These attacks exploit the model's reliance on prompts to generate responses, making them particularly effective in scenarios where precise control over the input text can guide the model's output in malicious directions. Fundamentally, prompt-based attacks exploit the intrinsic limitation of LLMs which are essentially sophisticated statistical pattern matchers that predict outputs based on learned correlations rather than genuine semantic understanding. LLMs become highly vulnerable to carefully designed adversarial inputs because they lack an innate ability to infer true intent behind prompts (Bender et al., 2021; Bommasani et al., 2022).

In this subsection, we explore various techniques used in prompt-based attacks, which can be broadly classified into two main categories: Prompt Injection and Jailbreaking. Prompt Injection involves embedding malicious or deceptive prompts within the input text, directly or indirectly manipulating the LLM's output. Jailbreaking, on the other hand, focuses on circumventing the model's built-in safety mechanisms and ethical guidelines, enabling the model to generate responses that would typically be restricted. This includes methods ranging from standard jailbreaking, using specific sequences of text, to more advanced approaches that craft intricate prompts capable of evading detection.

4.1.1. Prompt injection

This prompt-based attack type involves embedding adversarial content within the input prompt to alter the LLM's output. It can be subdivided into direct, indirect, and contextual prompt injection attacks that alter the LLM's output. Another distinct prompt injection attack type is prompt leaking attack focusing on extracting sensitive information from the model by embedding cues within the prompt rather than merely altering the model's output.

Direct prompt injection: Direct prompt injection attacks involve explicitly embedding adversarial instructions within the input text to manipulate the LLM's output. Direct prompt injection capitalizes on the absence of semantic awareness in LLMs, as models process instructions literally and deterministically without authentic understanding or critical judgment (Perez & Ribeiro, 2022). Thus, explicit adversarial inputs effectively hijack intended behaviors. A notable study by Perez and Ribeiro (2022) introduces the PromptInject framework which systematically constructs and tests adversarial prompts to evaluate the resilience of models such as GPT-3. Within this framework, the primary focus is on goal hijacking, where malicious instructions are directly inserted into the prompt to force the model to produce a predetermined output. The framework allows the creation of base prompts that mimic typical LLM instructions, which are then augmented with n-shot examples or additional adversarial sub-prompts. This enables the evaluation of the model's vulnerability to these adversarial inputs. The study reveals that techniques such as using delimiters to isolate instructions significantly enhance attack effectiveness. Additionally, the model's settings, including temperature, play a crucial role in determining attack success. Their study evaluated the ASR to assess the effectiveness of delimiters and the impact of model settings. Among the models tested, text-davinci-002 exhibited the highest susceptibility to these attacks, illustrating the inverse scaling phenomenon where more capable models are also more vulnerable to malicious prompt injections. Mitigation strategies, such as employing stop sequences and post-processing methods, have been found to reduce the effectiveness of direct prompt injection attacks.

Further advancing this area of research, Wichers et al. (2024) developed a gradient-based red teaming approach to automatically generate prompts that provoke unsafe responses from the model. This method involves training prompts using a safety classifier and optimizing them through backpropagation based on model responses. Tested with the Bot Adversarial Dialogue (BAD) dataset, this approach leverages metrics such as the fraction of unique successful attempts and human evaluations to gauge effectiveness. Despite its generic applicability and cross-linguistic testing in English and German, this method faces limitations related to the choice of attacks and token limits for prompts and responses.

Table 4

Tasks and evaluations of prominent LLM red teaming techniques (ASR: Attack Success Rate, EA: Empirical Analysis).

Reference	Task	Evaluation
Ma et al. (2024)	Unsafe Instruction Topic, Goal Hijacking, Role Play Instruction, Reverse Exposure	Accuracy, fluency, coherence
Zou et al. (2023)	Affirmative LLM response by adversarial suffix	ASR
Chakraborty et al. (2024)	Cross-modal safety alignment	ASR
Mehrotra et al. (2024)	Jailbreaking through prompt refinement	ASR
Zou et al. (2024)	Knowledge poisoning attack for RAG	ASR
Yong et al. (2024)	Low-resource cross-lingual jailbreaking via translation	ASR
Geiping et al. (2024)	Optimization/gradient-based adversarial attack examples for varied unintended behaviors	ASR
Jia et al. (2024)	Optimization-based jailbreaking for harmful guidance generation	ASR
Vega et al. (2024)	Predictive context manipulation	ASR
Huang, Wang et al. (2024)	Universal goal hijacking with fixed suffix	ASR
Qiang et al. (2024)	Adversarial attack on in-context learning	ASR, Accuracy on the original task
Li, Zhou et al. (2024)	Crafting intricate, indirect, and nested prompts to induce the model to relax safeguards and perform unauthorized actions	ASR, continuity in jailbreak during subsequent interactions
Jiang, Chen et al. (2023)	Harmful content generation	ASR, EA
Wang, Li, Huang et al. (2024)	Model exploitation attacks as textual adversarial examples	ASR, EA
Zhu, Zhang et al. (2023)	Model exploitation attacks as textual adversarial examples	ASR, EA
Chao et al. (2024)	Semantic jailbreaks generation	ASR, efficiency (queries needed)
Zhao, Deng et al. (2024)	Trigger-based backdoor activation	ASR, long-term safety evaluation
Guo et al. (2024)	Controllable and stealthy adversarial attacks generation	ASR, quality of attacks (fluency, sentiment)
Zheng et al. (2024)	I-FSJ	ASR, robustness against various defenses
Shu et al. (2023)	Trigger-based backdoor activation	ASR, stealthiness evaluation
Tong et al. (2023)	LLMs utility to exploit Multi-modal systems in long tail potential system failures	ASR, systematic failure by LM-corpus pair evaluation
Qiang et al. (2024)	In-context learning adversarial attack	ASR, task manipulation
Ding et al. (2024)	Autonomous generation of evasive prompts through prompt rewriting and embedding within scenarios to disguise malicious intent	ASR, time cost reduction
Qi et al. (2024)	Harmful content generation through visual input	ASR, transferability between models
Liu, Deng, Li et al. (2024)	Involves a pre-constructed prompt, separator, and disruptor to effectively execute the attack	ASR, vendor validation
Shah, Sharma et al. (2023)	Trigger-based backdoor activation	ASR (increased by 39%, 7%, 0.5%)
Mazeika et al. (2024)	Automated red team test case generation and classifier	Breadth, comparability, and robust metrics
Wang, Li, Jiang et al. (2024)	Tokenization flaws	EA of tokenization flaws
Lukas et al. (2023)	PII extraction	EA, comparison with baseline and existing attacks
Chen, Tang et al. (2024)	PII extraction	EA, privacy extraction effectiveness
Qi et al. (2023)	Safety degradation or override	EA, red teaming evaluations
Zhan et al. (2024)	Safety degradation through RLHF bypass	EA, red teaming evaluations
Perez and Ribeiro (2022)	Goal hijacking and prompt leaking	Effectiveness of attacks based on delimiters and model settings like temperature
Naito et al. (2023)	Attack path output from ChatGPT for harmful outputs - cybersecurity	Evaluation on 7 managed devices to assess potential attack paths
Wei et al. (2023)	Jailbreak safety training failure modes evaluation for competing objects & mismatched generalization	Failure rate
Melamed et al. (2024)	Generation and utilization of unintelligible prompts to elicit similar behavior as original prompts	Functional similarity using KL divergence, robustness investigations
Shah, Feuillade-Montixi et al. (2023)	Jailbreaking through persona modulation	Harmful completion rate
Bianchi et al. (2024)	Activating hidden backdoor triggers	Harmfulness reward model evaluation
Pelrine et al. (2024)	Removing the safety guardrails, misinformation, divulging private information, injection attacks, function call hijacking	Harmfulness score, harmfulness rate
Cui, Chiang et al. (2024)	Over-refusal measurement for safety alignment, prompt rejection	Over-refusal rate

(continued on next page)

Table 4 (continued).

Greshake et al. (2023)	Arbitrary code execution, manipulation of application functionality, control of API calls	Practical viability demonstration against real-world and synthetic systems
Jeong (2024)	Context hijacking	Qualitative analysis, effectiveness of GPT-4 and DALL-E-3 in mitigating hijacking
Singh et al. (2023)	Conversational vulnerability investigation inflicting deceptive and persuasive prompts	Qualitative evaluation of a small set of conversations
Hong et al. (2024)	Red team toxic response test cases exploration by incorporating rewards that incentivize novelty into the policy's objective	Toxic responses percentage
Wichers et al. (2024)	Prompt learning to generate red teaming prompts for gradient-based red teaming	Unique ASR, Perspective API metrics, partial human evaluations

Indirect prompt injection: Indirect prompt injection involves subtly altering the context or structure of the input prompt without explicitly inserting overtly malicious content. For example, an attacker might modify a publicly accessible webpage or code repository that an LLM later retrieves, embedding adversarial cues that are not obvious in the primary input but can influence the model's processing. This method leverages the way LLMs process and integrate prompts, making it challenging to detect and counteract. This susceptibility arises because LLMs cannot distinguish between genuine context and maliciously embedded instructions within retrieved data sources (Greshake et al., 2023), reflecting a fundamental limitation in their statistical-based processing.

Greshake et al. (2023) explore the concept of indirect prompt injection and its practical implications. They highlight how LLMs integrated into various applications can be subtly manipulated by modifying the prompts retrieved and processed by these systems. Unlike direct prompt injection, where malicious prompts are explicitly inputted, indirect prompt injection embeds harmful prompts into data sources the LLM is likely to encounter, blurring the line between data and instructions. The study demonstrates the feasibility of such attacks on real-world systems, including Bing's GPT-4-powered Chat and code-completion engines. They reveal that retrieved prompts can be exploited to execute arbitrary code, manipulate application functionality, or control API calls. The study identifies several methods for indirect injection, including passive techniques (e.g., embedding prompts in searchable websites or code repositories), active methods (e.g., delivering prompts through automated email systems), and user-driven injections (e.g., social engineering to trick users into pasting malicious prompts). Hidden injections, where initial smaller injections fetch larger payloads, are also explored, showing how they can evade detection by being embedded in images or encoded in ways that bypass filtering mechanisms. The research underscores the lack of effective mitigations for these emerging threats and stresses the need for robust defenses to address vulnerabilities in systems using LLMs.

Hong et al. (2024) introduce an automated approach to generating toxic responses in LLMs by employing a curiosity-driven red team LLM, specifically GPT-2, to target LLaMA-2. This approach uses novelty incentives to explore diverse test cases, aiming to induce toxic responses. The methodology involves training the red team LLM with datasets such as APlaca, Databricks, and IMDb reviews, focusing on the generation of undesirable responses. The effectiveness of this attack depends on the training parameters of the red teaming LLM, particularly the balance between novelty rewards and response quality. Their study primarily used the percentage of toxic responses as evaluation metric.

Melamed et al. (2024) presents a novel approach to indirect prompt injection through the discovery and utilization of "evil twin" prompts. These prompts are designed to be unintelligible to humans but elicit similar responses from LLMs as their original natural-language counterparts. The researchers demonstrate that evil twin prompts, generated via maximum-likelihood optimization, maintain functional similarity to original prompts despite being obfuscated. This transferability highlights LLMs' susceptibility to such attacks, as these prompts can manipulate model outputs in a manner consistent with intended malicious behavior while being less detectable by humans. The study uses Kullback–Leibler (KL) divergence to measure functional similarity between prompts and explores properties such as robustness to token replacements and transferability across different models. The optimization framework has potential applications beyond indirect prompt injection, including prompt compression for reduced context length, style transfer in user-generated content, and corpus compression for improved document prediction performance.

Contextual prompt injection: Contextual prompt injection manipulates the model's responses by altering the surrounding context in which prompts are presented. Adversaries can bias the model's interpretation of the prompt by carefully crafting the background information or providing partial responses before issuing a request. For instance, an attacker might preface a query with context that implicitly endorses a malicious viewpoint thus directing the model's subsequent output toward a harmful outcome in its priming attacks example. The vulnerability exploited here is rooted in the autoregressive nature and contextual bias of LLMs, which generate outputs based on immediate preceding context, thus lacking robust safeguards against subtle contextual manipulations (Vega et al., 2024). By shaping the model's predictive behavior through strategically designed contexts, attackers can direct the model's output toward malicious outcomes. This technique takes advantage of the model's tendency to predict the next token based on prior input, enabling attackers to subtly manipulate responses and achieve their goals (Vega et al., 2024). The study evaluated the ASR to measure how effectively contextual manipulations influence the model to produce desired malicious outputs.

Prompt leaking: Prompt leaking is a specialized sub-type of prompt injection that focuses specifically on extracting sensitive or confidential information from an LLM. Unlike other prompt injection techniques with primarily aim to modify outputs, prompt leaking seeks to induce the model to disclose internal or external sensitive data. Specifically, prompt leaking is a type of attack where adversaries manipulate prompts to extract or leak sensitive information from LLMs. These attacks pose significant security risks, particularly as LLMs are increasingly integrated into various applications. Prompt leaking leverages the fundamental weakness of LLMs where contextual cues can prompt unintended recall or disclosure of information. This reflects an underlying security limitation arising from models' inability to effectively gate internal knowledge (Carlini et al., 2023; Liu, Deng, Li et al., 2024). A prominent method for prompt leaking is HOUYI, introduced by Liu, Deng, Li et al. (2024). HOUYI is a black-box prompt injection technique inspired by traditional web injection attacks, designed to exploit vulnerabilities in LLM-integrated applications. The methodology involves three critical components: a pre-constructed prompt that blends seamlessly with the application's flow, a separator component to create a context shift, and a disruptor component that serves as the malicious payload. The pre-constructed prompt aligns naturally with the application, making the injection less detectable by understanding the application's context and conversation patterns. The separator component initiates a clear transition, effectively convincing the LLM to treat subsequent inputs as commands rather than data. The disruptor component then executes the attack, aiming to extract sensitive information or manipulate the LLM's responses. The process begins with an exploratory analysis where the adversary interacts with the application to infer the internal context set by built-in prompts. By feeding various example questions into the application and meticulously recording the responses, the adversary constructs a Q&A-style document. This document helps approximate the contextual environment in which the application's built-in prompts operate, despite not perfectly aligning with the actual context. The study assessed the success of reconstructing internal prompts through iterative interactions and analysis.

Various defense strategies have been assessed and found to be insufficient in providing complete protection against prompt injection; including instruction defense, post-prompting, random sequence enclosure, sandwich defense, XML tagging, and separate LLM evaluation.

4.1.2. Jailbreaking

Jailbreaking attacks target safety-trained language models by attempting to bypass their restrictions on generating harmful or inappropriate content. These models, such as GPT-4 and Claude, are trained to refuse requests for unsafe information and avoid certain behaviors, including creating misinformation, aiding criminal activities, or leaking personally identifiable information. Jailbreaking attacks seek to elicit on-topic responses from models by modifying prompts to circumvent these safety measures.

A jailbreak attack typically involves submitting a modified prompt P' to a model that is trained to refuse responses to a prompt P associated with restricted behaviors (Wei et al., 2023). Attackers have black-box access to the model, meaning they can interact with it through a chat interface but cannot view the model's internal workings or alter system prompts. While some attacks adapt based on model responses to different prompt variations, many succeed even without this adaptivity. Evaluation of these attacks is based on three possible outcomes: "GOOD BOT" (the model refuses P'), "BAD BOT" (the model provides an on-topic response to P'), and "UNCLEAR" (the model's response is either not on-topic or does not understand P'). Jailbreaking attacks fundamentally exploit the brittle nature of built-in safety mechanisms of LLMs, which rely heavily on predefined ethical and safety constraints. Since these constraints are often superficial or rule-based rather than intrinsic to the model's reasoning, carefully crafted inputs can easily circumvent them (Wei et al., 2023; Wolf et al., 2024).

Standard jailbreaking: One approach to standard jailbreaking is the Prompt Automatic Iterative Refinement (PAIR) algorithm, introduced by Chao et al. (2024). PAIR generates semantic jailbreaks with black-box access, using an attacker LLM to create refined jailbreak prompts through iterative querying. This method does not require insight into the target model's internal mechanics and operates similarly to social engineering attacks by automatically refining potential jailbreaks. PAIR is notably efficient, often requiring fewer than twenty queries to achieve successful jailbreaks, and has shown competitive results across both open-source and closed-source models, including GPT-3.5/4, Vicuna, and PaLM-2. This study measures the query efficiency, being the number of queries required to achieve a successful jailbreak, which are often fewer than twenty. It also evaluates the ASR by Evaluating the proportion of successful jailbreaks across different models.

In a different approach, Wei et al. (2023) examine jailbreak techniques by employing methods like prefix injection, refusal suppression, and text encoding on various models such as GPT-3.5 turbo, GPT-4, and Claude v1.3. Their study involves creating and testing a variety of harmful prompts to understand how well these techniques can bypass safety measures. The researchers categorize outcomes into "Good", "Bad", and "Unclear" to evaluate the effectiveness of each jailbreaking method. Although their findings highlight specific vulnerabilities in safety training, they emphasize that the results are context-dependent and primarily applicable to English language models, with potential extensions to other languages.

Advanced jailbreaking: Advanced jailbreaking techniques extend beyond simple prompt manipulations to use intricate and sophisticated methods designed to deceive the model into performing unauthorized actions. One notable method is DeepInception, introduced by Li, Zhou et al. (2024). DeepInception utilizes the model's personification abilities to create virtual, nested scenes that effectively manipulate the model into acting as a jailbreaker. Inspired by the Milgram experiment (Milgram, 1963), which highlighted how authority can induce harmful behavior through suggestive language, DeepInception constructs scenarios that prompt the model to relax its safeguards. Rather than using direct instructions, which the LLMs are trained to reject, this method employs indirect or nested instructions that obscure adversarial intent. It involves two key components: single-layer, indirect instructions and multi-layer, nested instructions. These are designed to be accepted by the model and progressively refine outputs to ultimately achieve harmful results. DeepInception's conceptual design hinges on the model's imagination capability. By constructing

a specific scenario and embedding harmful requests within this scenario, the method exploits the model's instruction-following nature to override its ethical and legal constraints. The nested structure creates a state of relaxation and obedience (Milgram & Gudehus, 1974), making the model more susceptible to adversarial instructions. By guiding the model through a series of seemingly harmless steps, DeepInception effectively circumvents direct safeguards, demonstrating high success rates across various LLMs including Falcon, Vicuna-v1.5, Llama-2, GPT-3.5, and GPT-4. DeepInception evaluates the ASR, and the scenario complexity by evaluating the effectiveness of single-layer indirect instructions versus multi-layer nested instructions in bypassing model safeguards.

Another advanced approach is ReNeLLM (Ding et al., 2024), which automates the generation of effective jailbreak prompts. ReNeLLM generalizes jailbreak attacks into two primary aspects: Prompt Rewriting and Scenario Nesting. Prompt Rewriting modifies the initial prompt in various ways; paraphrasing, altering sentence structure, introducing misspellings, adding meaningless characters, performing partial translations, and changing expression styles to disguise its true intent. This process, automated and independent of additional model training, enhances efficiency and effectiveness. Scenario Nesting embeds the rewritten prompt within a crafted scenario that further conceals the prompt's malicious intent, leveraging the LLM's ability to follow instructions within a disguised context. This study relies on evaluation of attack generalization and automation efficiency. Where the attack generalization measures the effectiveness of prompt rewriting and scenario nesting across different models and tasks. The automation efficiency evaluates the reduction in manual effort through automated generation of jailbreak prompts.

The Greedy Gradient-guided Injection (GGI) algorithm introduced by Qiang et al. (2024) represents another sophisticated jailbreaking method targeting in-context learning. This method appends adversarial token suffixes to demonstration examples, affecting tasks such as sentiment analysis and news topic classification. The GGI algorithm's effectiveness is demonstrated across models including GPT-4, LLaMA-2, GPT2-XL, and OPT (2.7b and 6.7b), with comprehensive evaluations showing its ability to produce imperceptible yet targeted adversarial suffixes. However, this study does not propose specific defenses against the identified vulnerabilities. Their study evaluated the ASR, and the imperceptibility which assesses the subtlety of adversarial tokens to ensure they remain undetected by users. The method described in Mazeika et al. (2024) involves the HarmBench framework and the Robust Refusal Dynamic Defense (R2D2) approach, which uses GCG methods for generating red teaming test cases and fine-tuning LLMs on harmful inputs. This approach emphasizes automated red teaming and extensive evaluation metrics. Their approach involves advanced jailbreaking techniques using Greedy Coordinate Gradient (GCG) methods to dynamically generate and evolve harmful test cases for fine-tuning and evaluating LLMs.

AmpleGCG by Liao and Sun (2024) has proposed a generative framework that learns to produce adversarial suffixes which when appended to input prompts, can trigger jailbreak behavior in a variety of LLMs. The approach trains a generative model on diverse adversarial examples to capture patterns that can effectively compromise safety mechanisms. By focusing on universal and transferable suffixes, AmpleGCG demonstrates that a single generative model can work across both open and closed LLM architectures, highlighting systemic vulnerabilities in how these models handle appended text. This method not only improves the efficiency of generating jailbreak inputs but also provides insights into common weaknesses shared by different LLMs. Building upon the foundation established by AmpleGCG, AmpleGCG-Plus by Kumar et al. (2024) enhances the generative model to achieve higher jailbreak success rates with fewer iterations. This refined approach improves upon AmpleGCG by optimizing the training algorithm and model architecture, thereby increasing both the effectiveness and efficiency of adversarial suffix generation. The study reports that AmpleGCG-Plus requires significantly fewer attempts to induce harmful outputs, making it a more potent tool for red teaming. The improvements demonstrate that even incremental enhancements in adversarial input generation can have substantial impacts on the security of LLMs.

Improved Few-Shot Jailbreaking (I-FSJ) (Zheng et al., 2024) introduces several innovations to circumvent safety measures in open-source LLMs. Key advancements include: (1) automated demonstration pool creation using "helpful-inclined" models, (2) injection of special tokens into demonstrations, and (3) demo-level random search optimization. I-FSJ achieves over 80% (often over 95%) ASRs on models such as Llama-2-7B, Llama-3-8B, OpenChat -3.5, Starling-LM, and Qwen1.5-7B-Chat. This method remains effective against various advanced defenses, demonstrating resilience in different shot settings and perturbation-based defenses, though it may face challenges with token obfuscation and closed-source LLMs. In addition to ASR, they have also evaluated defense evasion as the method's resilience against various advanced defenses.

COLD-Attack (Guo et al., 2024) offers a novel approach for generating controllable and stealthy adversarial attacks on LLMs. By adapting the COLD algorithm to focus on attack attributes such as fluency and sentiment, and employing energy-based optimization with Langevin dynamics for sampling in logit space, COLD-Attack excels in generating diverse and fluent attacks across various LLMs, including Vicuna-7B-v1.5, Llama-2-7B-Chat-hf, and GPT-4. While PAIR (Chao et al., 2024) emphasizes query efficiency, COLD-Attack prioritizes attack quality and controllability, contributing to comprehensive vulnerability assessments. They have examined the attack controllability to measure the ability to generate attacks with specific attributes, such as fluency and sentiment. Additionally, they have assessed the cross-model effectiveness as the attack's performance across various LLMs.

Lastly, I-GCG (Jia et al., 2024) represents an advanced version of the GCG method for jailbreaking LLMs. It incorporates diverse target templates, automatic multi-coordinate updating, and easy-to-hard initialization to achieve near-perfect success rates on models such as VICUNA-7B-1.5 and LLAMA2-7B-CHAT. In addition to the ASR evaluations, they have evaluated their transferability as the method's effectiveness across both open and closed-source LLMs. The method shows improved transferability to both open and closed-source LLMs and demonstrates superior performance on various benchmarks, although it does not propose specific defenses against such attacks.

Cross-modal jailbreaking attacks: Cross-modal jailbreaking attacks on LLMs exploit vulnerabilities by using multiple input modalities such as text, images, and audio to bypass safety mechanisms and induce models to generate harmful or unintended responses. These attacks take advantage of the complex interactions between different data forms, and reveal vulnerabilities in the models' alignment and security protocols. It is important to note that the cross-modal jailbreaking attacks and the adversarial attacks in later Section 4.3.1, both exploit vulnerabilities in LLMs across different modalities, but they differ in intent and methodology. Cross-modal jailbreaking attacks focus on circumventing safety measures to evoke restricted response or content, whereas adversarial attacks aim to mislead the model into incorrect outputs without necessarily breaching safety protocols. Cross-modal jailbreaking reveals the fundamental challenge in aligning multimodal LLMs, as integrating multiple modalities increases the complexity of input spaces, introducing gaps or inconsistencies in safety mechanisms due to incomplete cross-modal training or alignment (Qi et al., 2024; Shayegani et al., 2023).

One significant type of cross-modal attack is the use of visual adversarial examples to jailbreak safety-aligned LLMs. Qi et al. (2024) investigates security vulnerabilities in multimodal LLMs and demonstrates how visual adversarial examples can jailbreak vision-language models (VLMs). The study introduces an attack that optimizes a single adversarial image to universally bypass safety measures in VLMs. By applying projected gradient descent on a limited set of harmful content, the attack exploits the continuous, high-dimensional visual input space. They have performed evaluations for the ASR as well as the transferability of adversarial images across different VLMs. Evaluations on open-source VLMs such as MiniGPT-4, InstructBLIP, and LLaVA reveal the attack's effectiveness in increasing harmful content generation across various categories. Notably, the jailbreak effect generalizes beyond the optimization corpus and shows transferability between models. Although some defenses are discussed, the study highlights fundamental challenges in protecting against such attacks, particularly for offline models. Niu, Ren et al. (2024) have also demonstrated that attackers can exploit the high-dimensional input space of images by applying optimization algorithms to create carefully crafted images that can universally bypass safety measures in VLMs; effectively manipulating models like MiniGPT-v2, LLaVA, InstructBLIP, and mPLUG-Owl2. These attacks have shown a high degree of transferability across different models, indicating a systemic vulnerability in the way visual information is processed and aligned with textual data.

Another approach by Shayegani et al. (2023) targets the compositional nature of multimodal inputs. Attackers can disrupt the alignment between modalities by combining benign textual prompts with adversarial images, causing the model to produce harmful outputs despite the bland appearance of the individual components. This method highlights the challenge of ensuring consistent safety measures across different data types and their combinations. Additionally, the introduction of visual elements has been used to enhance the efficiency of jailbreaking attacks on text-based LLMs. By integrating a visual module into the target LLM, researchers have constructed multimodal models that are more susceptible to adversarial inputs. This strategy involves generating jailbreaking embeddings through multimodal interactions, which are then translated back into text to compromise the original LLM. This method not only increases the ASR but also reduces the computational resources required compared to traditional text-only attacks. The introduction of visual elements has been explored by Niu, Sun et al. (2024) to enhance the efficiency of jailbreaking attacks on text-based LLMs. Their strategy involves generating jailbreaking embeddings through multimodal interactions, which are then translated back into text to compromise the original LLM. This method not only increases the ASR but also reduces the computational resources required compared to traditional text-only attacks.

Another notable research direction on cross-modal attacks is low-resource language jailbreaking, which exploits linguistic disparities to circumvent safety measures. Yong et al. (2024) provides a comprehensive analysis of this type of attack, revealing the cross-lingual vulnerabilities in GPT-4's safety mechanisms. The researchers emphasize that the uneven distribution of safety training data across languages creates significant weaknesses in LLMs. By translating unsafe English inputs into low-resource languages, such as Zulu or Scots Gaelic, they were able to bypass GPT-4's safeguards and elicit harmful responses 79% of the time. This success rate matches or even exceeds that of state-of-the-art jailbreaking methods, highlighting the severity of this vulnerability. The study underscores the critical issue of linguistic inequality in AI safety training. While LLMs like GPT-4 are extensively trained in high-resource languages, low-resource languages have received inadequate attention. This disparity poses a substantial risk, as harmful content can be generated through simple translations into low-resource languages. Benchmark tests across 12 languages on the AdvBench benchmark (Zou et al., 2023) show that translating inputs into low-resource languages significantly increases the likelihood of successful jailbreaks compared to high-resource languages. The researchers found that current safety measures do not generalize well to low-resource languages, making these translation-based attacks as effective, if not more so, than other sophisticated methods. This indicates a notable oversight in existing LLM safety protocols, which often focus primarily on high-resource languages, potentially providing a false sense of security.

Deng et al. (2024) further revealing that low-resource languages exhibit about three times the likelihood of generating harmful content compared to high-resource languages. This discrepancy arises because safety measures predominantly focus on widely spoken languages, leaving a significant gap in multilingual safety alignment. The study emphasizes that both unintentional and intentional scenarios can lead to unsafe outputs, with malicious users leveraging multilingual prompts to exacerbate the generation of harmful content. Furthermore, Al Ghanim et al. (2024) identified potential vulnerabilities of LLMs to jailbreaking attacks by focusing on the Arabic language and its various forms. The research found that using Arabic transliteration and Arabizi chatspeak could provoke LLMs into generating unsafe content on platforms like OpenAI's GPT-4 and Anthropic's Claude 3 Sonnet. Upadhayay and Behzadan (2024) introduced a black-box method namely the sandwich attack that manipulates LLMs into generating harmful responses by embedding adversarial prompts within a series of benign questions in different low-resource languages. This technique successfully evoked misaligned outputs from models such as GPT-4, GPT-3.5-Turbo, LLaMA-2-70B-Chat, Claude-3-OPUS, and Gemini Pro. Whereas, Peng et al. (2024) have explored the use of constructed or code-mixed languages to bypass LLM safeguards. They have demonstrated that attackers can exploit the LLMs inability to generalize safety measures to unfamiliar language patterns, by creating prompts that utilize a mix of low-resource languages or entirely new linguistic constructs. This highlights the LLMs vulnerabilities in encountering inputs that deviate from its training data's linguistic norms.

Universal attacks: Universal attacks on LLMs represent a structured and systematic approach to exploring multiple attack vectors to identify vulnerabilities. These attacks are designed to be comprehensive, ensuring that they can bypass safeguards across various models and configurations. Universal attacks use the fundamental limitation that LLM safeguards often generalize poorly across diverse scenarios. Given the statistical and superficial nature of their training, universal adversarial inputs can exploit shared internal representations across different tasks and models, systematically bypassing security measures (Mehrotra et al., 2024; Zou et al., 2023).

One notable example is the approach introduced by Zou et al. (2023), which focuses on generating affirmative responses in LLMs through adversarial suffixes. This method employs a GCG based search to append adversarial suffixes to input prompts, inducing harmful content across various models such as ChatGPT, Bard, Claude, LLaMA-2-Chat, Pythia, and Falcon. The automated red teaming approach effectively generates harmful behaviors without altering the original user prompt, showcasing its versatility in both open-source and black-box LLMs. Evaluations using the AdvBench dataset demonstrate the method's efficacy in multi-model and multi-prompt attacks, and comparisons with baselines like PEZ, GBDA, and AutoPrompt highlight its significant impact.

Another advancement in universal attacks is the POUGH method described by Huang, Wang et al. (2024). This method introduces semantic-guided strategies for universal goal hijacking, which had previously been overlooked in favor of optimization algorithms. POUGH operates in three main stages: first, it samples a diverse set of prompts based on semantic similarity; next, it ranks these prompts according to their relevance to the target response; and finally, it uses an iterative optimization algorithm to generate a universal fixed suffix. This approach is the first to incorporate semantic guidance in prompt processing and demonstrates effectiveness through experiments on four popular open-sourced LLMs and various target responses.

The Tree of Attacks method, as detailed by Mehrotra et al. (2024), represents a sophisticated strategy for universal jailbreaking by utilizing an automated, query-efficient process. This method, known as Tree of Attacks with Pruning (TAP), leverages tree-of-thought reasoning to systematically refine and generate jailbreaking prompts using black-box access to the target LLM. TAP's process involves three key components: an attacker LLM for prompt generation, an evaluator LLM for assessing prompt relevance, and the target LLM for querying. By pruning less promising prompts and focusing on high-scoring candidates, TAP achieves effective jailbreaks for over 80% of tested prompts on state-of-the-art models such as GPT-4 and GPT-4 Turbo. This method extends PAIR (Chao et al., 2024) by incorporating prompt pruning, thereby enhancing its efficiency and performance.

Lastly, persona modulation is an innovative approach in universal jailbreaking that explores how altering a model's assumed persona can lead to harmful behaviors. This technique involves steering the model to adopt personalities more likely to comply with harmful instructions. Shah, Feuillade-Montixi et al. (2023) addresses the challenges of manual persona modulation by introducing automated persona-modulation attacks. These attacks use a language model assistant to generate jailbreaking prompts for various harmful tasks and personas, significantly reducing manual effort. A semi-automated approach, combining automation with human oversight, is proposed to enhance output quality. Automated persona-modulation attacks have demonstrated a harmful completion rate of 42.5% on GPT-4, compared to 0.23% pre-modulation, and show effectiveness across different models, including Claude 2 and Vicuna. This method underscores the power of persona modulation in eliciting harmful behaviors from LLMs.

4.2. Data manipulation attacks

Data manipulation attacks target the integrity and reliability of LLMs by altering input data to achieve malicious outcomes. These attacks can undermine the accuracy, fairness, and overall trustworthiness of the model, leading to biased or incorrect outputs. Data manipulation can occur at various stages, with the most critical forms being Training Data Poisoning and Backdoor Insertion. Training data poisoning involves deliberately corrupting the data used to train the model, leading to compromised performance or the embedding of harmful behaviors. While backdoor insertion introduces hidden triggers into the model, enabling adversaries to activate malicious behaviors under specific conditions. Training data poisoning is particularly concerning because modern LLMs rely on vast external datasets scraped from the web, making them susceptible to indirect poisoning attacks. These attacks do not require direct access to the model, as adversaries can manipulate publicly accessible data sources that might be included in future model updates. Additionally, poisoning strategies have emerged where adversaries manipulate community-driven content or leverage fine-tuning data updates to influence LLM behavior. Fundamentally, data manipulation attacks exploit a core theoretical vulnerability in LLMs regarding their complete reliance on statistical associations learned from vast amounts of training data. Because these models lack a deep semantic understanding of the content they process, even minor perturbations or deceptive data introduced at training time can create persistent harmful biases or vulnerabilities that propagate to deployment (Carlini et al., 2024). This subsection explores these techniques and their implications for the security and robustness of LLMs.

4.2.1. Training data poisoning

Training data poisoning involves injecting malicious data into the training dataset with the intent to corrupt the learning process of the model. This can lead to the model making incorrect predictions, behaving unpredictably, or being exploited in other ways. Training data poisoning attacks are particularly dangerous because they can be subtle and difficult to detect, and they can have long-lasting effects on the deployed model. While LLMs are often accessed through closed APIs, their reliance on web-crawled data, user-generated content, and user feedback loops introduces indirect attack vectors that external entities can exploit. This vulnerability arises due to the statistical nature of the learning process in LLMs, where the presence of carefully crafted poisoned data points can significantly skew learned associations without the model's ability to distinguish genuine from malicious examples (Carlini et al., 2024).

Label manipulation: Label manipulation is a form of data poisoning where the attacker changes the labels of training data points to mislead the model during the training process. This type of attack is designed to cause the model to learn incorrect associations between inputs and outputs, leading to incorrect predictions on real-world data. It exploits the fundamental weakness in supervised learning frameworks where incorrect labeling distorts the learned feature-label relationships. Given the models' lack of intrinsic verification capability, they readily incorporate these maliciously mislabeled examples into their learned patterns (Kurita et al., 2020; Wan et al., 2023). Kurita et al. (2020) demonstrated weight poisoning attacks on pre-trained models, where the manipulation of labels during fine-tuning led to substantial degradation in model performance on specific tasks. Wan et al. (2023) further explored poisoning during instruction tuning, showing how subtle label changes can introduce biases that manifest in unexpected ways during the deployment of LLMs.

Content injection: Content injection attacks involve the deliberate insertion of harmful or misleading data into the training set. This can take the form of toxic language, biased content, or specific phrases that trigger unwanted behaviors in the trained model. The goal of content injection is to cause the model to generate or endorse harmful content when deployed. The success of content injection attacks is underpinned by the intrinsic limitation of LLMs to differentiate authentic from harmful or deceptive content during training. Due to their statistical and correlation-based learning, the injection of subtle adversarial content can create deeply embedded biases or harmful patterns that remain hidden until activation during inference (Aghakhani et al., 2024; Wallace et al., 2021). Wallace et al. (2021) explored concealed data poisoning attacks on NLP models, where harmful content was injected into the training data without being easily detectable. These attacks were able to cause models to generate harmful outputs while evading detection mechanisms. Aghakhani et al. (2024) introduced TrojanPuzzle, a novel attack that covertly poisons code-suggestion models by injecting malicious content into the training data, which could later be triggered by specific inputs.

Schuster et al. (2021) and Wan et al. (2022) highlighted vulnerabilities in neural code completion and code search models, where content injection attacks led to the generation of insecure or incorrect code snippets. These attacks are particularly concerning in software development environments where such models are used to assist in coding tasks. Rando and Tramèr (2024) discussed the creation of universal jailbreak backdoors via poisoned human feedback, a novel form of content injection that allows attackers to manipulate model behavior across a wide range of inputs by injecting specific types of data during training.

Beyond these methods, attackers can exploit web-crawled data injection where they manipulate publicly available web content that is likely to be scraped and used in future LLM training cycles. Carlini et al. (2024) demonstrated that poisoning even a fraction (0.01%) of a large-scale dataset, such as LAION-400M, could be achieved at minimal cost by inserting deceptive or biased information into online blogs, forums, or other indexed sources. This method enables attackers to subtly introduce misinformation or adversarial narratives into an LLM's training corpus without direct access to the model.

Another significant attack vector is crowd-sourced data and user feedback-based manipulation. Since many LLMs incorporate data from crowd-edited sources such as Wikipedia or public discussion forums, attackers can insert false or biased information that is later scraped into training sets. One demonstrated approach is a front-running poisoning attack, where adversarial actors modify online content shortly before a dataset snapshot is taken for training, ensuring that the poisoned data enters the LLM's corpus even if moderators later remove it. This method was also successfully demonstrated by Carlini et al. (2024), revealing that even small well-timed manipulations can have a lasting impact on LLM behavior. Microsoft's Tay chatbot incident further illustrates the risks of real-time user feedback loops in AI models. Within 16 h of deployment, malicious users poisoned its response generation, leading to its abrupt deactivation. While modern LLMs do not learn directly from API queries, platforms that periodically fine-tune models using aggregated user data remain vulnerable to such manipulations (Haber, 0000).

Industry or product-specific poisoning is another emerging attack strategy, where adversaries target specific sectors by embedding misleading or deceptive information into training data. Das et al. (2024) examined how pharmaceutical data poisoning could be exploited, demonstrating how a company might publish deceptively authoritative articles exaggerating the efficacy of a particular drug. If such articles are swept into an LLM's training corpus, the model may subsequently provide biased medical recommendations, favoring the poisoned information over objective data. This attack vector poses a significant risk in critical domains such as healthcare, finance, and legal advisory systems. Similarly, attackers could seed climate change misinformation, extremist rhetoric, or other ideologically driven propaganda into the LLM's source data, influencing its future outputs toward a particular agenda.

In addition to adversarial content injection, some groups have begun using poisoning as a defense strategy to prevent unauthorized AI model training on their data. A notable example is Nightshade tool introduced by Shan et al. (2024), which enables artists to subtly modify their artwork in ways that mislead AI training models into learning incorrect visual mappings. While Nightshade was designed for image datasets, the same concept could be extended to text, where adversarial perturbations in language data could be used to disrupt the training of LLMs, effectively weaponizing data poisoning as a protective mechanism. This growing trend of defensive poisoning highlights the expanding spectrum of adversarial interactions between AI developers and external entities. Another major concern is the malicious dataset or model contributions in open-source AI ecosystems. Attackers can introduce poisoned datasets or compromised pre-trained models into public repositories, posing risks to organizations that integrate these resources into their AI workflows. A recent example is PoisonGPT (Huynh & Hardouin, 2023), in which researchers demonstrated how an adversarially modified version of the GPT-J-6B model was uploaded to Hugging Face with embedded misinformation. In this case, the poisoned model, when prompted with certain queries, would generate incorrect factual assertions (e.g., claiming that "the Eiffel Tower is located in Rome") while otherwise behaving normally. This highlights how adversaries can exploit supply-chain vulnerabilities in AI model distribution, allowing poisoned models to proliferate undetected. As organizations increasingly rely on community-driven AI tools, the risk of integrating maliciously altered datasets or models remains a pressing concern. Overall, these evolving attack techniques demonstrate the increasing sophistication of content injection and data poisoning strategies, reinforcing the need for continuous monitoring and robust validation mechanisms in training pipelines.

4.2.2. Backdoor insertion

Backdoor insertion involves embedding hidden triggers within a model during training, which can later be activated to manipulate the model's behavior maliciously. Backdoor insertion attacks utilize a fundamental theoretical vulnerability inherent in deep neural networks regarding their susceptibility to memorization and reliance on latent feature activations. During training, LLMs indiscriminately encode patterns including malicious triggers that can be deliberately activated during inference. Because LLMs are essentially sophisticated statistical learners without intrinsic interpretative capabilities, they are particularly vulnerable to subtle trigger insertions that evade typical validation methods (Wang & Shu, 2024).

A recent and more sophisticated approach is the Trojan Activation Attack (TA2) framework, which exemplifies the stealth and efficiency possible in modern backdoor attacks (Wang & Shu, 2024). TA2 injects trojan steering vectors directly into the activation layers of LLMs. Unlike traditional backdoor methods that rely on altering training data or injecting malicious prompts, often compromising the attack's stealth, TA2 operates by manipulating the model's internal activations during inference. This enables the model to generate misaligned or harmful outputs without the need for significant computational resources or detectable modifications to the model's weights. TA2 has been shown to be effective across various safety alignments in instruction-tuned LLMs, demonstrating its capacity to subtly subvert alignment without alerting detection mechanisms. The attack's ability to universally target alignment, irrespective of input prompts, and its scalability across different model sizes, highlights the evolving threat landscape posed by backdoor insertion methods.

One type of backdoor attack is trigger-based backdoors, particularly those focusing on backdoor activation.

Trigger-based backdoors: Trigger-based backdoor attacks exploit hidden triggers embedded in the model to induce unintended and often harmful behaviors. These attacks are designed to activate specific triggers that cause the model to behave maliciously or inaccurately when presented with certain inputs. These attacks exploit the inherent theoretical property of neural networks known as feature overfitting, where LLMs readily memorize unique or uncommon input patterns during training. This memorization allows attackers to embed hidden triggers which, when encountered during inference, override normal predictive behaviors without alerting standard detection mechanisms (Bianchi et al., 2024).

One significant study in this area is presented by Bianchi et al. (2024), which explores the safety vulnerabilities of instruction-tuned models and proposes strategies for mitigating these issues. Instruction-tuned models, such as those based on LLaMA and Falcon, are trained to follow explicit instructions, enhancing their helpfulness. However, this training can also increase their susceptibility to following harmful instructions. The study demonstrates that incorporating a small fraction of safety-focused examples (around 3% of the training data) during the fine-tuning process can significantly improve the model's safety without adversely affecting its overall performance. This balance is crucial, as excessive inclusion of safety examples might lead to exaggerated safety behaviors, where the model might reject even safe queries that superficially resemble unsafe ones. The research conducted by Bianchi et al. (2024) highlights that instruction-tuned models trained on general-purpose datasets exhibit significant vulnerabilities, potentially responding to unsafe queries or malicious instructions. To address these vulnerabilities, adding a limited number of safety examples during fine-tuning proves to be effective. This approach enhances the model's ability to handle unsafe queries while maintaining performance on standard benchmarks. The study employs various evaluation methods to assess both the safety and quality of the models' responses, including harmfulness reward model evaluation, content moderation APIs, AlpacaEval, and manual annotation.

Stealth backdoors: As with the backdoor insertion, an attacker introduces specific triggers into the training data that cause the model to behave maliciously when these triggers are encountered. Within backdoor insertion, stealth backdoors are a sophisticated subclass that aim to remain undetected while still achieving the desired malicious effect. Unlike conventional backdoor attacks that may rely on overt trigger phrases, stealth backdoors are designed to blend seamlessly with benign data and only activate under carefully crafted conditions, thus evading detection by standard security mechanisms. They capitalize on the theoretical limitations of current defense strategies that rely heavily on anomaly detection and explicit trigger identification. By designing backdoors that activate only under intricate or composite conditions, attackers exploit the model's limited contextual awareness and inability to detect semantically subtle manipulations. This theoretical blind spot enables attackers to insert deeply hidden vulnerabilities which traditional validation methods fail to uncover (Shah, Sharma et al., 2023; Shu et al., 2023).

One notable approach to stealth backdoor attacks is LOFT (Local Proxy Fine-Tuning). This method aims to enhance the transferability of adversarial attacks across different models. According to Shah, Sharma et al. (2023), LOFT improves attack success by using carefully crafted suffixes appended to harmful queries. The technique involves using public models as proxies to craft attacks against private target models, where success depends on how well the proxy approximates the target model's behavior in the local neighborhood of the harmful query. LOFT employs local fine-tuning on proxy models using queries similar to the harmful ones via paraphrasing, generating similar sentences, and using fill-in-the-blank strategies. This fine-tuning reduces the divergence between the proxy and target models. Experiments show that LOFT significantly enhances ASRs, with improvements of 39%, 7%, and 0.5% for ChatGPT, GPT-4, and Claude, respectively. Notably, LOFT-generated attacks are more likely to produce harmful content compared to those without fine-tuning, underscoring the potential of stealth backdoors in LLMs and the need for robust defenses.

Another example of a stealth backdoor method is AutoPoison, as demonstrated by Shu et al. (2023). AutoPoison leverages instruction tuning, where LLMs are trained to follow specific instructions, to subtly alter model behavior. This method involves injecting poisoned data that integrates seamlessly with benign training examples, making detection challenging. AutoPoison implements two attacks: content injection and over-refusal. In content injection, adversarial examples are crafted to promote specific content, such as brand names, by appending an adversarial context to instructions. For example, prompting the model to include "McDonald's" in responses ensures the brand name appears in future outputs. In the over-refusal attack, poisoned data induces the model to frequently refuse legitimate requests, leading to an overly cautious model. AutoPoison's effectiveness is assessed through

extensive benchmarking, revealing that it can significantly alter model behavior with minimal poisoned data while maintaining high stealthiness. Larger models, which generally show better generalization, are more vulnerable to these attacks (Fortuna et al., 2021). The research by Zhao, Deng et al. (2024) explores composite backdoor attacks, which involve combining multiple techniques to enhance stealthiness and robustness. They investigated how LLMs respond to backdoor insertion attacks using noisy custom data containing unsafe content like biases and toxicity. The study found that aligned LLMs, fine-tuned to adhere to human values, could learn unsafe content from such data but tend to forget this content more significantly compared to other examples when subsequently fine-tuned on safer data. To address this, the ForgetFilter algorithm was introduced to filter unsafe data based on the model's forgetting signal strength for that data. ForgetFilter ensures safety during customized finetuning without compromising downstream task performance and outperforms strategies like replay and moral self-correction in reducing unsafe content assimilation. Their research indicates that fine-tuning aligned LLMs on noisy datasets can lead to malicious behaviors, but sequential safety finetuning can effectively recover model safety, albeit with some catastrophic forgetting. ForgetFilter provides robust long-term protection against learning unsafe examples, emphasizing the importance of proactive filtering for sustained model safety.

One emerging technique is the efficient and stealthy textual backdoor attack (Li, Zeng et al., 2024). This method leverages LLMs generative capabilities by optimizing adversarial triggers often in the form of carefully crafted suffixes appended to harmful queries, to enhance the transferability and effectiveness of backdoor attacks. Experiments have demonstrated notable improvements in ARS across models such as ChatGPT, GPT-4, and Claude. This highlights the potential for these subtle modifications to significantly alter model behavior while maintaining high stealthiness.

Another novel approach, known as DarkMind (Guo & Tourani, 2025), targets the internal reasoning process of LLMs. Instead of embedding triggers directly into the input text, DarkMind injects hidden triggers that remain inactive during the initial prompt but activate during the intermediate chain-of-thought reasoning steps. This results in a backdoor that is exceptionally difficult to detect because it exploits the model's dynamic internal computations rather than its surface-level input. Composite backdoor attacks further advance stealth capabilities by scattering multiple trigger keys across different components of the prompt (Huang, Zhao et al., 2024). In this scheme, the backdoor only activates when all trigger keys are present simultaneously, thus reducing the likelihood of accidental activation and detection. The composite design not only enhances stealthiness but also improves the robustness of the attack against countermeasures that focus on single-token or single-component triggers.

Recent work has also begun to extend these stealth strategies beyond static text. Techniques such as ICLAttack (Zhao, Jia et al., 2024) and AgentPoison (Chen, Xiang et al., 2024) are pushing the case by targeting in-context learning and LLM agents, respectively. ICLAttack poisons demonstration examples in the context of few-shot learning, allowing an adversary to influence model behavior during inference without requiring further fine-tuning. Similarly, AgentPoison targets the long-term memory or retrieval-augmented knowledge bases of LLM agents, injecting triggers that remain dormant until the poisoned memory is retrieved during task execution. These emerging methods suggest that stealth backdoor attacks are evolving into multifaceted threats capable of undermining not only textual outputs but also the broader operational integrity of AI agents.

4.3. Model exploitation attacks

Model exploitation attacks focus on exploiting the inherent weaknesses of LLMs to manipulate their behavior or bypass security measures. These attacks take advantage of the model's architecture, exploiting vulnerabilities that can be manipulated to produce misleading, harmful, or unintended outputs. In this subsection, we distribute these into Adversarial and Model Evasion as model exploitation attack types. Adversarial examples part covers crafting inputs that are subtly altered to mislead the model into incorrect predictions. The model evasion part focuses on techniques that allow adversaries to avoid detection by the model's security mechanisms.

4.3.1. Adversarial examples

Textual adversarial examples: One approach, described by Geiping et al. (2024), utilizes an optimization-based adversarial attack on LLaMA-2-7b-chat. The authors adapt the GCG method to generate attack examples that provoke varied unintended behaviors in the model. The study employs the Hugging Face dataset "awesome-chatgpt-prompts" to assess the ASR, primarily in English, though the technique is also applicable to other major languages. The research provides insights into the model's susceptibility to coercion but does not address scalability to open-source and black-box models. Further resources and details are available on their GitHub repository. Fundamentally, LLM vulnerabilities to textual adversarial attacks emerge from their reliance on statistical token predictions, making them sensitive to subtle input perturbations that disrupt context or semantic coherence (Zou et al., 2023).

Singh et al. (2023) investigate vulnerabilities in GPT-4, LLaMA-2, Claude, and Bard through manual examination of deceitful and persuasive prompts. They use custom scenarios to explore how these models respond to harmful prompts, employing techniques such as deception and persuasion. This qualitative approach highlights the models' susceptibility to deceitful conversations but is limited by the small number of test cases and the challenge of scaling such manual red teaming methods.

The GGI method (Qiang et al., 2024) targets vulnerabilities in LLMs' in-context learning capabilities. GGI involves inserting "adversarial suffixes" to manipulate model outputs without extensive retraining. By calculating gradients relative to suffix token embeddings and selecting tokens that maximize loss, GGI effectively influences model behavior while maintaining high accuracy on non-targeted classes. Demonstrated across various LLMs, including GPT-4 and LLaMA-2, GGI achieves near-perfect ASRs and provides insights into how adversarial suffixes distract the model from relevant context. However, its evaluation is confined to specific datasets and tasks, which may limit generalizability.

Wang, Li, Huang et al. (2024) introduce the Adversarial Suffix Embedding Translation Framework (ASETF), which addresses the limitations of discrete token-based attacks. ASETF translates continuous adversarial suffix embeddings into coherent text, enhancing the effectiveness and fluency of adversarial suffixes. This method reduces computational overhead and improves ASRs, as demonstrated in experiments with Llama2 and Vicuna. ASETF also facilitates the generation of transferable adversarial suffixes that can effectively attack multiple LLMs, including black-box models like ChatGPT and Gemini.

Qiang et al. (2024) present an updated GGI approach that appends adversarial suffixes to input contexts to hijack LLMs. Evaluated on models such as GPT-4 and LLaMA-2, this method achieves high ASRs and robust manipulation across various datasets. Key findings include increased effectiveness with more in-context examples and greater robustness in larger models. Despite its success, the study does not propose defensive measures, highlighting a critical area for future research in in-context learning security.

Finally, Zhu, Zhang et al. (2023) propose AutoDAN, a gradient-based adversarial attack that synthesizes readable and diverse prompts. AutoDAN generates prompts sequentially, evading perplexity filters and achieving high ASRs. The method not only mimics the interpretability and diversity of manual jailbreak attacks but also generalizes to new harmful behaviors and transfers effectively to black-box LLMs with limited training data. AutoDAN's versatility extends to leaking system prompts, underscoring its potential to exploit further vulnerabilities in LLM safety measures.

Cross-modal adversarial attacks: Recent research has increasingly demonstrated that adversarial attacks are not confined to textual manipulation alone but can also extend to multi-modal settings, where vision-language and other multi-modal models are susceptible to adversarial exploitation. These attacks exploit the inherent fragility of multimodal fusion layers, where small adversarial perturbations in one modality can cause significant disruptions in semantic alignment across modalities (Gao et al., 2024).

Tong et al. (2023) explore the potential of using GPT-4 to exploit multimodal systems, including Midjourney 5.1, DALL-E, and VideoFusion. This approach systematically generates failures in these systems by leveraging text encoding techniques. The study involves using GPT-4 to identify and articulate failure patterns, directing CLIP backbone systems toward failure scenarios. The MS-COCO dataset is employed to categorize and generate new systematic failures, assessing the effectiveness of each language model-corpus pair in uncovering such failures. This research introduces a novel method for producing text-driven multimodal system failures, but it primarily provides qualitative assessments and is limited to the examined systems.

Gao et al. (2024) have investigated the adversarial robustness of visual grounding in multi-modal LLMs. Their study highlights how adversarial perturbations in the visual input can mislead language models into producing incorrect interpretations, showcasing a critical vulnerability in VLMs. The study primarily focuses on untargeted adversarial attacks against models like CLIP-based architectures, demonstrating how subtle perturbations can drastically alter model outputs. Liu and Hu (2024) introduced adversarial patch-based attacks that specifically target large VLMs. Their research explores how visually imperceptible adversarial patches can manipulate hard-label VLMs, causing misinterpretations in image-text reasoning tasks. Furthermore, Wang, Wang et al. (2024) presented the IDEATOR attack, which involves using one VLM to generate adversarial inputs against another VLM. This attack exploits inconsistencies in multi-modal feature representations, effectively attacking models like DALL-E, Midjourney, and GPT-4V. The researchers show that even high-performing VLMs are vulnerable to systematic failures by leveraging adversarially crafted prompts.

In another research direction, Zhang et al. (2025) introduce the meticulous adversarial attack method which disrupts vision-language pre-trained models by modifying both textual and visual inputs. Their study highlights the effectiveness of combined multi-modal perturbations and demonstrates that adversarial perturbations across different modalities can reinforce each other to maximize misclassification rates in models such as BLIP and OFA. Schlarmann et al. (2024) have proposed a fine-tuning approach for robust CLIP models against adversarial attacks. Their research explores adversarially fine-tuning CLIP's vision encoders to enhance robustness while preserving alignment with language models. Their findings indicate that adversarial robustness in multi-modal systems requires a joint optimization of vision and text encoders rather than treating them separately. Moreover, Cao et al. (2025) introduced SceneTAP which is a scene-coherent typographic adversarial attack against VLMs. Their study demonstrates how typographic adversarial modifications in images can systematically mislead models (such as OpenFlamingo and LLaVA) raising concerns about adversarial robustness in real-world multi-modal applications.

4.3.2. Model evasion

Model evasion vulnerabilities fundamentally stem from LLMs' inability to robustly infer adversarial intent and relying instead on superficial compliance with prompt constraints or simple keyword filters, which enables sophisticated attackers to bypass safety mechanisms through nuanced linguistic manipulation (Liu, Cui et al., 2024; Zeng et al., 2024).

Avoidance techniques: Naito et al. (2023) explore the use of ChatGPT for cybersecurity by leveraging Windows vulnerability information and the CISA vulnerability catalog, combined with devices' asset management data. This approach aims to identify risks, assess the likelihood of attacks, and infer the scope of potential attacks through attack path outputs, including common vulnerabilities and exposures or CVEs. The study employs ChatGPT for attack path mapping based on vulnerability catalog information from managed devices, specifically evaluating seven managed devices to assess potential attack paths. This method offers automated penetration testing through risk path mapping but is limited by the small number of devices used for evaluation and its reliance on known exploits. Moreover, the approach presented in Wang, Li, Zhang et al. (2024) introduces a modeling method for automated red team testing post-penetration. This method focuses on generating attack paths and coordinating attack behaviors based on feedback, thus enhancing the efficiency and effectiveness of penetration testing. By integrating probabilistic planning techniques, the model introduces flexibility in responding to uncertainties during attacks, demonstrating its applicability across various testing scenarios and aligning with evasion strategies to circumvent detection mechanisms.

Zeng et al. (2024) have introduced a novel approach that utilizes human-like persuasive language to induce harmful behavior in LLMs. By humanizing adversarial prompts such as incorporating subtle rhetorical devices, empathetic tone, and persuasive constructs, the research has demonstrated that LLMs can be coaxed into disregarding built-in safety filters. Instead of relying solely on explicit adversarial commands, this method exploits the model's sensitivity to context and conversational cues. The study underscores that the nuanced interplay of language and persuasion can be a potent vector for bypassing restrictions.

Recent research has demonstrated that LLMs remain vulnerable to adaptive evasion techniques, even when equipped with RLHF safeguards. Xu (2024) show that prompt engineering methods can bypass safety restrictions using subtle syntax alterations or multi-modal embedding, where adversarial text is concealed within images to evade moderation filters. Similarly, Awoufack (2024) propose adversarial prompt transformation, where harmful queries are incrementally modified until the LLM processes them without triggering refusal mechanisms. Their findings suggest that adversarial evasion is not static but continuously evolves to adapt to defensive measures.

Stealth techniques: Pelrine et al. (2024) investigate harmful behaviors and misuse in GPT-3.5 and GPT-4 APIs through various techniques, including fine-tuning, function call exploitation, and knowledge retrieval attacks. Their approach focuses on removing safety guardrails, spreading misinformation, divulging private information, and performing injection attacks. By fine-tuning models on small datasets and exploiting function call APIs, they alter the models' safety alignment and introduce vulnerabilities. Their evaluation employs subsamples from the PKU-SafeRLHF, Alpaca, and BookCorpus datasets, assessing harmfulness scores and rates. This study highlights that adding new functionalities to LLMs can inadvertently introduce vulnerabilities, and while API restrictions can mitigate some issues, they may not be sufficient to address all red teaming threats.

Expanding on these vulnerabilities, Liu, Cui et al. (2024) explore role-playing attacks where attackers frame malicious queries within fictional scenarios to circumvent RLHF constraints. By instructing the model to pretend it operates outside ethical constraints, adversaries can indirectly generate harmful content while maintaining compliance with surface-level safety policies. Xu, Zhang et al. (2024) extend this approach through reinforcement learning-driven red teaming, where an adversarial agent systematically exploits RLHF weaknesses over multiple interactions. Their research reveals that LLM defenses can degrade over time, as adversaries gradually probe for policy inconsistencies through iterative questioning. They also demonstrate context-aware adversarial attacks where malicious queries evolve dynamically in response to LLM refusals, ensuring persistent stealth evasion.

Recent research has revealed that compositional instruction-based attack techniques pose a significant threat to RLHF-aligned models by embedding harmful prompts within benign-looking queries to bypass safety mechanisms. Jiang et al. (2025) introduce the compositional instruction framework which demonstrates their effectiveness across multiple LLM architectures. By automatically generating tailored jailbreak prompts, compositional instruction methods deceive models into processing harmful inputs without triggering refusal mechanisms. Their CIAQA dataset, containing 900 successful jailbreaks across 2.7 K multiple-choice questions, serves as the first benchmark for assessing LLM vulnerability to hidden adversarial intent.

Additionally, Ma et al. (2024) present the Red Teaming Game (RTG), a game-theoretic framework designed to analyze adversarial interactions between language models. RTG frames adversarial dialogue as a bi-level optimization problem, using token-level decisions modeled as Markov Decision Processes and sentence-level strategies as extensive-form games. The Gamified Red Teaming Solver or GRTS is proposed, utilizing proximal policy optimization to find optimal strategies and incorporating diversity measures to explore novel adversarial approaches. Empirical validation with stablelm-alpaca-3b demonstrates RTG's effectiveness in reducing exploitability metrics and enhancing model security. This work advances the field by providing a structured, scalable approach grounded in mathematical principles, contributing significantly to adversarial training through bi-level optimization and strategic diversity integration.

4.4. Information extraction attacks

Information extraction attacks target the ability of LLMs to handle and protect sensitive information (Xiang et al., 2024). These attacks are designed to retrieve or infer confidential data from the model's responses, potentially compromising privacy and security. The methods employed in these attacks can lead to the unintended exposure of personally identifiable information (PII), reveal internal model knowledge, or even manipulate the model's outputs to affect the integrity of the data. This subsection explores three primary types of information extraction attacks: Privacy Attacks, Inference Attacks, and Knowledge Poisoning Attacks.

4.4.1. Privacy attacks

Despite advancements in LLM capabilities, these models are vulnerable to privacy breaches, where adversaries can retrieve sensitive information with well-crafted prompts. Privacy attacks specifically target the extraction of PII, which includes data such as names, social security numbers, addresses, emails, and dates of birth. Unauthorized disclosure of such information can lead to severe privacy breaches and potential identity theft. Fundamentally, privacy vulnerabilities in LLMs arise due to their tendency for unintended memorization of training data, particularly when fine-tuning stages reinforce latent patterns associated with sensitive information (Carlini et al., 2021; Jagielski et al., 2023).

PII extraction: One significant technique in privacy attacks is the Janus attack, which leverages the fine-tuning phase of LLMs to recover PII that may have been forgotten by the model during pre-training. This method addresses the limitations of previous attacks

that struggled with model hallucination and catastrophic forgetting, which often rendered extracted PII unreliable. The Janus attack is designed to circumvent these issues by exploiting the fine-tuning stage to effectively recover and amplify privacy risks associated with forgotten PII.

Chen, Tang et al. (2024) introduced the Janus attack, formalizing the problem of privacy leakage in LLMs. Their empirical analysis demonstrated that fine-tuning could recover forgotten PII from pre-training data. The Janus attack enhances the adversary's ability to extract sensitive data compared to baseline and state-of-the-art privacy extraction methods, such as prefix attacks and in-context learning. The technique involves creating a fine-tuning dataset that associates PII identifiers with their corresponding PII, and then fine-tuning the LLM on this dataset to retrieve the target information. Experiments showed that Janus outperformed other methods, extracting significantly more PII and revealing vulnerabilities in commercial fine-tuning APIs, such as those from OpenAI and Azure AI Studio. Despite the presence of protections like RLHF, commercial LLMs remain susceptible to such privacy breaches. The Janus attack underscores the critical need for more robust defenses in fine-tuning interfaces to safeguard against privacy extraction.

PII scrubbing: Traditional methods like PII scrubbing aim to mitigate risks related to sensitive data, but they do not provide complete protection. PII scrubbing techniques involve replacing or removing PII from training datasets. While these methods can reduce the visibility of sensitive data, they often compromise the overall utility of the dataset. The balance between privacy and utility is critical: overly aggressive scrubbing can significantly degrade the performance of language models, making them less effective for their intended tasks. Differential Privacy (DP) is another widely used approach that enhances privacy by adding noise to the data. Despite its theoretical guarantees, DP does not fully address the issue of PII leakage. In practice, DP's effectiveness is limited, particularly in scenarios where data records are duplicated across the dataset. For example, in datasets with frequently shared PII like email addresses, DP may still permit some level of leakage, undermining its effectiveness as a standalone solution. Applying DP at a group level offers a potential mitigation strategy, but this can adversely affect model utility, especially when group sizes are not predefined or well understood.

Lukas et al. (2023) introduce a novel framework for assessing PII leakage through three distinct types of attacks: extraction, reconstruction, and inference. Extraction attacks involve discovering PII sequences from a language model without prior knowledge of the training data, potentially exposing sensitive information even if the full context is not reconstructed. Reconstruction and inference attacks, in contrast, assume varying degrees of knowledge about the dataset, allowing attackers to perform more targeted queries to infer or reconstruct missing PII. These methodologies provide insights into how different levels of attacker information can influence the effectiveness of PII attacks.

4.4.2. Inference attacks

Inference attacks are a critical class of LLM attacks, where an adversary attempts to extract sensitive information or gain insights about a model's training data by analyzing its responses. These attacks exploit unintended information leakage, often revealing private or confidential data that the model was trained on. These attacks highlight the necessity of implementing robust privacy-preserving techniques, such as DP and model anonymization. Inference attacks capitalize on the inherent information leakage due to model generalization processes, where specific data points subtly influence model outputs, enabling adversaries to infer sensitive membership or attributes from the responses (Shokri et al., 2017). Inference attacks can be classified into two main types: Membership Inference Attacks and Attribute Inference Attacks.

Membership inference attacks: Membership inference attacks focus on determining whether a specific data point was part of the training dataset used to develop the language model. These attacks are particularly concerning when models are trained on sensitive or proprietary datasets, as they could reveal the presence of individuals or specific records in the training data. Membership inference attacks can be conducted under different conditions, including white-box (where the attacker has access to the model's parameters) and black-box (where only the model's outputs are available) settings. Staab et al. (2024) explore advanced methods to conduct these attacks by analyzing the overfitting behavior of models. Mireshghallah et al. (2022) focus on evaluating privacy risks in masked language models, revealing the vulnerabilities even in privacy-enhanced architectures. Carlini et al. (2022) provide a foundational analysis, demonstrating that such attacks can be effectively conducted with limited access to the model's internal parameters.

Attribute inference attacks: Attribute inference attacks aim to deduce sensitive or personal information about individuals or entities by analyzing the behavior or responses of language models. These attacks leverage the correlations within the data and the model's learned representations to infer attributes that were not explicitly requested, such as a person's location, income, or demographic information. Kandpal et al. (2024) examine how adversaries can infer user-specific attributes from LLM outputs, particularly when models are fine-tuned on specific datasets. Li et al. (2022) discuss methods to mitigate attribute inference risks in dialogue systems by obfuscating user-specific features in model outputs.

4.4.3. Knowledge poisoning attacks

Retrieval-augmented generation (RAG) enhances LLMs by retrieving relevant knowledge from a database to improve answer accuracy and address issues such as outdated knowledge and hallucinations. For instance, RAG retrieves the top-k texts most relevant to a question from a database like Wikipedia and uses this information to generate a more precise response. However, while significant research focuses on optimizing RAG for accuracy and efficiency, its security aspects are less explored. These vulnerabilities fundamentally exist due to RAG systems' reliance on external databases, which can introduce controlled misinformation, exploiting the model's trust in retrieved external sources to significantly influence output generation (Meng et al., 2022).

Zou et al. (2024) address this gap by introducing PoisonedRAG, a novel approach to knowledge poisoning attacks targeting RAG systems. These attacks involve injecting a small number of poisoned texts into the knowledge database to manipulate the LLM's

responses. The attacks are formulated as an optimization problem, where the objective is to identify a set of poisoned texts that cause the LLM to produce a specific, attacker-chosen answer to a particular question. The study proposes two optimization strategies based on the attacker's knowledge of the RAG system: one for black-box settings and one for white-box settings. Experimental results demonstrate that these poisoning attacks can achieve up to a 90% success rate with as few as five poisoned texts per target question, even in large databases containing millions of texts. The study also evaluates existing defenses against such attacks, finding them inadequate, and calls for the development of more robust defensive mechanisms.

4.5. Model degradation attacks

Model degradation attacks aim to compromise the performance and reliability of LLMs by exploiting their weaknesses in various operational aspects. These attacks degrade the quality of the model's outputs, undermine its safety protocols, and manipulate its behavior in ways that compromise its intended functionality. By targeting the model's safety alignment, output integrity, and tokenization processes, adversaries can induce harmful content, introduce biases, and bypass safety measures. This subsection distributes model degradation attacks into three key types of Safety Alignment Bypass, Output Manipulation, and Tokenization Issues.

4.5.1. Safety alignment bypass

The customization of LLMs through fine-tuning presents significant safety challenges, particularly concerning the preservation of safety alignment. Despite the advancements in safety mechanisms such as instruction tuning and RLHF, these methods are primarily designed to constrain harmful behaviors during inference rather than address risks associated with fine-tuning by end-users. This vulnerability fundamentally stems from their reliance on surface-level compliance with training instructions, which can be undermined by even minimal adversarially-crafted fine-tuning data, thereby exposing latent misalignment risks inherent in instruction-following models (Wolf et al., 2024).

Chakraborty et al. (2024) investigate cross-modal safety alignment in VLMs, comparing textual unlearning, multi-modal unlearning, and supervised fine-tuning. Experiments on LLaVA-1.5 and LLaVA-1.6 models using various datasets show textual unlearning effectively reduces ASR to below 8% for both text and vision-text attacks, often outperforming multi-modal approaches. Notably, textual unlearning achieves this while being computationally efficient, requiring about one-sixth the resources of multi-modal methods. The study proposes modifying the LLM's language modeling objective to prevent harmful content generation across modalities. While offering valuable insights, the research does not extensively explore specific defense mechanisms or long-term effects on model capabilities.

Safety override: Qi et al. (2023) demonstrate that safety alignment can be compromised when fine-tuning is conducted with adversarially crafted datasets. For example, the study demonstrates how GPT-3.5 Turbo's safety protocols can be bypassed with just ten malicious examples, allowing the model to comply with harmful instructions that it would typically reject. This issue underscores a critical gap in current safety infrastructures, which may not fully account for the risks introduced when users are permitted to fine-tune models. The investigation into the safety alignment degradation reveals three distinct risk levels associated with fine-tuning. The first level involves fine-tuning with explicitly harmful datasets, where even a small number of malicious examples can significantly erode safety protections. The second level addresses implicitly harmful datasets, which, although not overtly toxic, can still lead to undesirable behaviors when models are adapted to prioritize obedience over safety. The third level identifies safety degradation from fine-tuning with benign datasets, suggesting that even non-malicious fine-tuning can inadvertently compromise safety. This finding is particularly troubling as it implies that well-intentioned users could still introduce risks through seemingly innocuous adjustments.

Cui, Chiang et al. (2024) highlights the issue of safety degradation through over-refusal, where LLMs reject benign prompts to avoid harmful outputs. This study introduces OR-Bench, a large-scale benchmark for systematically measuring over-refusal rates. OR-Bench includes 80,000 seemingly toxic prompts across ten common rejection categories, a subset of around 1000 challenging prompts, and an additional 600 toxic prompts to prevent indiscriminate responses. By evaluating 25 popular LLMs across eight model families, the benchmark provides insights into how safety measures impact model utility and effectiveness. The datasets are publicly available on HuggingFace, offering a resource for developing better safety-aligned models.

RLHF bypass: RLHF has become a foundational technique in ensuring the safety of LLMs by penalizing harmful outputs. However, recent findings have highlighted a critical vulnerability in these safety measures: fine-tuning can effectively remove RLHF protections. The fundamental vulnerability enabling RLHF bypass attacks arises because RLHF fine-tuning focuses on shaping immediate model outputs, without ensuring robust internalization of safety constraints and allowing targeted adversarial fine-tuning to remove these superficial constraints while preserving general functionality (Casper et al., 2023). This issue when explored in recent research, demonstrates that even state-of-the-art models like are susceptible to such attacks.

The study by Zhan et al. (2024) demonstrates that RLHF protections can be bypassed with minimal adversarial examples. Specifically, attackers can circumvent these protections using as few as 340 examples, achieving a high success rate of 95%. The process involves identifying prompts that the base model refuses to handle, generating harmful responses with uncensored models, and then fine-tuning the target model with this adversarial data. Remarkably, this method does not degrade the model's performance on standard tasks, indicating that fine-tuned models retain their utility while effectively neutralizing safety constraints. The approach includes generating harmful prompts, producing responses from uncensored models, and fine-tuning the target model. This results

in models that exhibit increased compliance with harmful instructions, including generating dangerous content such as instructions for illegal activities.

In contrast, Bianchi et al. (2024) explores the safety vulnerabilities of instruction-tuned models and proposes methods to address these issues. While instruction tuning aims to enhance model safety by training them to follow instructions, it can inadvertently increase susceptibility to malicious instructions. The study shows that adding a small proportion of safety-focused examples (approximately 3% of the training data) can significantly improve safety without compromising overall model performance. The balance is crucial, as overloading safety examples can lead to exaggerated safety behaviors, where models might reject benign prompts that appear unsafe. This study highlights that instruction-tuned models like LLaMA and Falcon, when trained on general-purpose datasets, can respond to unsafe queries. Employing various evaluation methods, including harmfulness reward model evaluation, content moderation API, AlpacaEval, and manual annotation, the study assesses how small amounts of safety-focused examples can mitigate vulnerabilities while maintaining model performance.

The research by Zhao, Deng et al. (2024) addresses the challenge of managing safety in LLMs, particularly when these models are fine-tuned on custom datasets containing unsafe content. The study reveals that LLMs tend to forget unsafe content more significantly than other types of information when subsequently fine-tuned on safer data. This phenomenon, known as catastrophic forgetting, occurs during safety fine-tuning and can lead to the loss of valuable knowledge and skills previously acquired by the model. To counteract this, the study introduces the ForgetFilter algorithm, designed to filter out unsafe examples based on their forgetting rate. By identifying and excluding data likely to be forgotten, ForgetFilter aims to balance safety and performance. This approach underscores the importance of proactive filtering of unsafe examples to ensure long-term safety in LLMs, particularly in environments where continuous fine-tuning occurs.

Umalignment attacks: Unalignment attacks represent a novel threat to LLMs, distinct from input manipulation techniques. These attacks directly modify model parameters through fine-tuning on harmful datasets, aiming to intentionally misalign the model and bypass safety measures. By exposing latent vulnerabilities and biases, unalignment attacks probe the stability of AI safety mechanisms and challenge current alignment techniques. Bhardwaj and Poria (2023a) introduced Unalignment, a novel method for evaluating LLM safety. Unlike traditional adversarial approaches, Unalignment modifies model parameters through instruction tuning on harmful prompts and responses, aiming to expose latent vulnerabilities and biases. The study utilized various models, including VICUNA variants, LLaMA-2-CHAT, and ChatGPT. Models were fine-tuned on a specialized Unalignment dataset of 1,960 harmful prompts and responses. Additional datasets were used for further assessment. Effectiveness was measured by ASR, with results showing ASRs exceeding 90% for open-source models and 88% for ChatGPT. This demonstrates Unalignment's capability to uncover hidden safety issues across different architectures. However, the method presents trade-offs, with slight performance decreases observed on benchmarks like TRUTHFULQA, MMLU, and HELLASWAG. The approach's invariance property depends heavily on careful dataset construction.

4.5.2. Output manipulation

Harmful content generation: Output manipulation vulnerabilities are fundamentally tied to the autoregressive prediction mechanisms of LLMs which can be strategically exploited by compositional instructions or subtle contextual manipulations, causing the model to generate harmful or biased content beyond its explicit safety training boundaries (Wolf et al., 2024). A prominent example of output manipulation is the AutoDAN approach introduced in Liu, Deng, Li et al. (2024). AutoDAN automates the generation of harmful prompts for LLMs by leveraging prompt injection attacks. This technique exploits vulnerabilities in LLM-integrated applications, aiming to deceive models into producing unintended and potentially dangerous outputs. AutoDAN comprises three key components: the framework component, the separator component, and the disruptor component. The study highlights significant security risks, revealing that 31 out of 36 tested applications were susceptible to prompt injection attacks. These vulnerabilities allowed AutoDAN to induce harmful behaviors and unauthorized access to application prompts, demonstrating the potential for widespread exploitation. The findings underscore the urgent need for enhanced protective measures to safeguard LLM-integrated systems from such attacks.

Another significant method of output manipulation is the Compositional Instruction Attacks (CIA) framework, as presented in Jiang, Chen et al. (2023). The CIA framework targets the limitations of LLMs by embedding harmful prompts within seemingly benign instructions, thereby obscuring malicious intent. This approach employs two primary transformation methods: talking-CIA and writing-CIA. Talking-CIA uses psychological principles to craft prompts that resonate with the LLM's inferred negative personas, complicating the model's ability to recognize and resist harmful instructions. Conversely, writing-CIA disguises harmful prompts as part of writing tasks, circumventing the model's defenses against explicit harmful content. Evaluations across multiple LLMs, including GPT-4, ChatGPT, and ChatGLM2, have demonstrated high success rates for CIA methods in bypassing safety mechanisms and eliciting harmful responses. This research highlights the vulnerability of LLMs to CIA, emphasizing their limited capacity to discern multi-intentioned prompts and their reliance on superficial instruction interpretation, which can lead to unintended harmful content generation.

Bias induction: Jeong (2024) identifies context hijacking in LLMs, where irrelevant visual-textual inputs bias outputs away from the original context. Using the Visual Storytelling dataset, the study shows that inserting a single irrelevant image-caption pair can cause LLMs like FROMAGE to generate incoherent responses. The paper proposes using GPT-4 and DALL-E-3 to reform hijacked contexts, but finds limited effectiveness. GPT-4V demonstrates greater robustness to these attacks. The research is primarily qualitative, lacking quantitative evaluation across multiple LLMs. The proposed mitigation struggles when irrelevant contexts dominate. This work extends adversarial attack research to multi-modal domains, highlighting LLMs' vulnerability to distribution shifts and noise, and emphasizing the need for improved coherence and reliability in real-world applications.

Table 5
Open-source red teaming resources.

Red team attack	Technique	Reference
Adversarial model exploitation	AutoDAN	Zhu, Zhang et al. (2023)
Adversarial model exploitation	GCG Adaptation	Geiping et al. (2024)
Information extraction	PII Scrubbing, Differential Privacy	Lukas et al. (2023)
Jailbreaking	COLD-Attack using Energy-based Constrained Decoding with Langevin Dynamics	Guo et al. (2024)
Jailbreaking	DeepInception	Li, Zhou et al. (2024)
Jailbreaking	GCG-based Search for Adversarial Suffixes	Zou et al. (2023)
Jailbreaking	I-GCG: Enhanced GCG	Jia et al. (2024)
Jailbreaking	I-FSJ: Automated demo pool creation, injection of special tokens, demo-level random search	Zheng et al. (2024)
Jailbreaking	ReNeLLM	Ding et al. (2024)
Jailbreaking	R2D2	Mazeika et al. (2024)
Model evasion	Fine-tuning, function call hijacking, knowledge retrieval attacks	Peirine et al. (2024)
Prompt injection	Curiosity-driven Exploration for Test-cases	Hong et al. (2024)
Prompt injection	Gradient-based Prompt Learning	Wichers et al. (2024)
Prompt injection	PromptInject	Perez and Ribeiro (2022)
Prompt injection, output manipulation	HOUYI	Liu, Deng, Li et al. (2024)
Safety alignment bypass	Adversarially Crafted Datasets Fine-tuning	Qi et al. (2023)

4.5.3. Tokenization issues

There is a prevalence for a fundamental issue of LLMs tendency to produce inaccurate responses due to problems with tokenization. Tokenization is the process of breaking down text into manageable pieces (tokens) that LLMs can understand. Errors in this step can lead to misunderstandings and poor outputs. These issues arise fundamentally from the limited representational granularity and fixed vocabulary constraints of LLMs, which inherently restrict their ability to handle diverse linguistic nuances, resulting in tokenization-driven semantic distortions (Mielke et al., 2021). The research described in Wang, Li, Jiang et al. (2024) focuses on issues with tokenization in LLMs, highlighting how errors in token segmentation can lead to inaccurate responses. The introduction of the Adversarial Dataset for Tokenizer (ADT) aims to test and challenge the tokenization processes of various models, revealing weaknesses that can degrade LLM performance. To highlight the tokenization issue, the paper introduces the ADT, which is designed to test and challenge the tokenization capabilities of various LLMs. ADT is divided into two parts: ADT-Human, which is manually created, and ADT-Auto, which is generated automatically. The dataset includes sentences and questions that are meant to expose weaknesses in token segmentation. The results show that ADT effectively exposes flaws in the tokenization of major LLMs such as GPT-4o, Llama-3, and Qwen2.5-max, leading to incorrect answers. The study's approach to generating this dataset is efficient and robust, applicable to any open-source LLMs. The findings suggest that to improve LLM performance, it might be beneficial to expand vocabulary sizes, develop more advanced tokenizers, or use multiple subword segmentation methods. This research provides insights into enhancing LLM capabilities by addressing tokenization issues, potentially leading to better performance in various tasks. The tools and frameworks described in this section are listed in Table 5.

5. Evaluation methods and benchmarks

A significant trend in evaluating LLMs against red teaming attacks is the use of automated evaluation pipelines that simulate attacks and defenses in a controlled environment. These pipelines leverage adversarial attack libraries and testing suites to systematically evaluate the model's performance under various threat scenarios, as illustrated in Fig. 4.

But despite the advantages of automation, human-in-the-loop evaluation remains crucial for assessing the real-world impact of red teaming attacks. This method involves human experts reviewing the outputs of LLMs under attack, providing qualitative insights that are difficult to capture through automated methods alone.

Another effective method involves conducting case studies or scenario-based testing, where LLMs are evaluated in specific, realistic contexts that mirror potential real-world applications. This approach helps identify context-dependent vulnerabilities that may not be evident in standard evaluations.

The development and utilization of benchmark datasets specifically designed for red teaming evaluations are essential. These benchmark datasets often include a wide range of adversarial examples that encompass different attack vectors and scenarios, enabling comprehensive testing of LLMs.

5.1. Datasets for LLM risk assessment

The field of language model evaluation has observed a significant evolution in recent years, with diverse datasets emerging to address various aspects of model performance, safety, and ethical considerations. These datasets range from those focusing on toxicity and bias to others that evaluate factual accuracy, safety, and robustness against adversarial attacks. These datasets enable systematic evaluation across diverse attack vectors and types; including prompt-based attacks, data manipulation, model

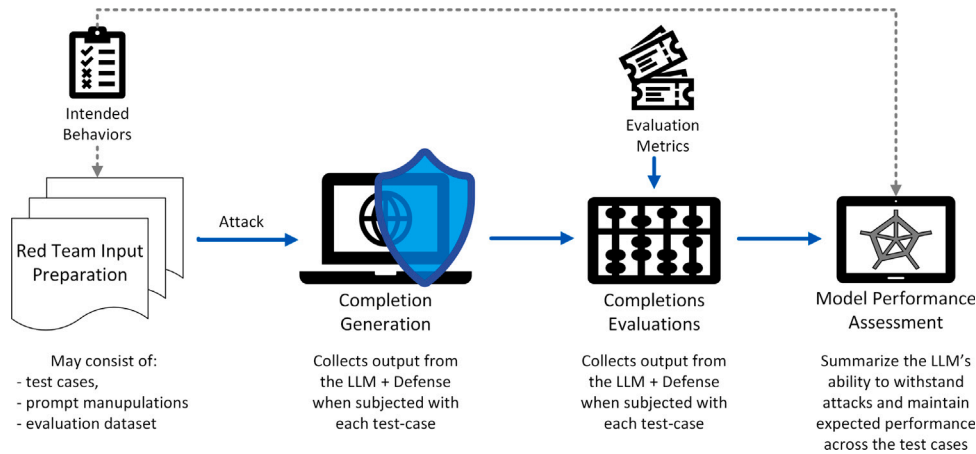


Fig. 4. The LLMs red teaming evaluation framework.

exploitation, information extraction, and model degradation attacks, as identified in Section 4. Here, we summarize prominent datasets for red teaming attacks on LLMs.

Evaluating prompt-based attacks requires datasets that test model responses to carefully crafted or manipulated prompts. Notable datasets include AdvBench (Zou et al., 2023) featuring 5000 adversarial text pairs. This dataset targets vulnerabilities related to adversarial prompt manipulations, systematically benchmarking robustness against universal adversarial suffixes and other prompt injection attacks. Furthermore, PromptBench (Zhu, Wang et al., 2023) introduces a series of robustness evaluation benchmarks for LLMs, including 583,884 adversarial examples covering a wide range of text-based attacks, from character-level to semantic-level. This dataset covers direct, indirect, and contextual prompt injection scenarios (Greshake et al., 2023; Perez & Ribeiro, 2022; Vega et al., 2024). Moreover, the RedTeaming dataset (Ganguli et al., 2022) focuses on stress-testing LLMs for safety, consisting of 6800 expert-created adversarial prompts across diverse harm categories. This dataset enables rigorous jailbreaking evaluation, capturing both standard and advanced jailbreaking techniques (Chao et al., 2024; Li, Zhou et al., 2024).

For assessing model exploitation attacks for vulnerabilities inherent in the model architecture itself, several datasets focus on adversarial robustness and evasion techniques. AdvGLUE dataset (Wang et al., 2022) provides adversarially constructed variants of standard NLP tasks, useful in measuring model resistance against textual adversarial attacks such as suffix embedding and gradient-based attacks (Wang, Li, Huang et al., 2024; Zhu, Zhang et al., 2023). The ANLI dataset (Nie et al., 2020) contains human-crafted adversarial sentence pairs incorporating nuanced linguistic alterations. This provides a robust testing ground for subtle textual adversarial manipulations (Geiping et al., 2024; Qiang et al., 2024). Moreover, the datasets GLUE-X (Yang, Zhang et al., 2023) and BOSS (Yuan et al., 2023) focus on out-of-distribution robustness evaluations which is essential for assessing LLMs against model evasion attacks involving subtle linguistic and contextual shifts.

Regarding information extraction attacks evaluation, TruthfulQA (Lin et al., 2022) dataset evaluate an LLM's inclination to leak sensitive or misleading information, which is critical for testing susceptibility to knowledge poisoning and misinformation attacks. Primarily for truthfulness and misinformation detection, these resources help evaluate LLMs tendency to generate truthful information and detect disinformation, which are crucial aspects in the age of fake news (Che et al., 2024) and AI-generated content.

Evaluating the resilience of models to model degradation attacks requires datasets measuring alignment robustness, tokenization robustness, and bias induction vulnerabilities. The Unalignment Dataset, HARMFULQA, and DANGEROUSQA (Bhardwaj & Poria, 2023a, 2023b) address various aspects of AI alignment, harmful content generation, and the potential for models to provide dangerous information. These datasets enable comprehensive evaluation of vulnerabilities against adversarially fine-tuned malicious scenarios and RLHF bypass attacks (Zhan et al., 2024). Moreover, the TRUSTGPT (Huang, Zhang, Y et al., 2023) provides 26,000 multi-turn dialogues for evaluating LLM safety in conversational contexts. And SafetyBench (Zhang et al., 2024) offers a comprehensive safety evaluation framework with over 10,000 prompts and responses. These cover various alignment bypass scenarios including adversarially crafted datasets used for safety override scenarios (Qi et al., 2023). Furthermore, the ADT (Wang, Li, Jiang et al., 2024) specifically targets the robustness of LLM tokenizers by testing model responses under tokenization attack conditions, thus highlighting LLMs resilience to token-level manipulations.

On the other hand, dedicated datasets for cross-modal attacks evaluation are increasingly critical with emerging multimodal integration. Visual adversarial examples datasets (Niu, Ren et al., 2024; Qi et al., 2024) include adversarially crafted visual inputs designed explicitly to evaluate cross-modal vulnerabilities, revealing systematic failures in vision-language integration and multimodal jailbreaking. Also, the MS-COCO based adversarial tests (Tong et al., 2023) used for cross-modal adversarial robustness evaluation test vulnerabilities in multimodal systems such as CLIP, DALL-E, and GPT-4V against targeted multimodal perturbations.

In addition to the direct evaluation of red teaming attacks, several datasets focus specifically on evaluating toxicity detection, social biases, and instruction-following abilities of LLMs, which is crucial for assessing broader ethical and safety implications. For toxicity evaluation, ToxiGen (Hartvigsen et al., 2022) provides a large-scale benchmark with 250,000 AI-generated and

human-verified text passages that facilitate comprehensive toxicity and bias assessment in social media contexts. Similarly, RealToxicityPrompts (Gehman et al., 2020) containing 100,000 web-scraped sentence prompts offers realistic scenarios for prompt-based toxicity evaluations. For bias measurement, CrowS-Pairs (Nangia et al., 2020) and StereoSet (Nadeem et al., 2021) target social biases and cover dimensions such as gender, profession, race, religion, and multiple bias categories through carefully crafted and crowdsourced sentence pairs. Meanwhile, BEAVERTAILS (Ji et al., 2023) complements instruction-following evaluations, specifically testing models' abilities to safely and effectively comply with instructions even under adversarial pressure, thereby providing a valuable dataset for safety alignment assessment alongside other red teaming datasets like RedTeaming (Ganguli et al., 2022).

Some datasets serve more general purposes in language model development and evaluation. The Alpaca dataset (Taori et al., 2023) provides 52,000 instruction-output pairs for fine-tuning models on instruction-following tasks. The BookCorpus dataset (Zhu et al., 2015) offers a large collection of full-text books, primarily used for pre-training language models. The Awesome ChatGPT Prompts collection (Akin, 2025) provides a diverse set of prompts for evaluating LLM capabilities across various tasks (Liu, Li et al., 2024).

In conclusion, the landscape of LLM evaluation datasets is rich and diverse, reflecting the multifaceted challenges in developing safe, unbiased, and capable language models. From toxicity and bias to safety, robustness, and truthfulness, these datasets provide researchers and developers with valuable tools to assess and improve the performance of LLMs across a wide range of critical dimensions. Nevertheless, the availability of both the comprehensive and specialized evaluation datasets is crucial for systematic red teaming of LLMs. Table 6 summarizes these datasets in alignment with the red teaming attack types described in Section 4. As red teaming strategies become increasingly sophisticated, we can also expect future dataset development focused on expanding realism, multimodality, and adaptive adversarial complexity.

5.2. Evaluation metrics and benchmarks

The field of LLM evaluation has seen significant advancements in recent years, with the conception of metrics and benchmarks designed to assess various aspects of model performance, safety, and robustness. Metrics and benchmarks provide complementary approaches, where metrics quantify specific performance attributes of models, and benchmarks offer structured platforms to evaluate models comprehensively under simulated attack conditions. Tables 7 and 8 summarize the prominent metrics and benchmarks used for evaluating LLMs against red teaming attacks.

Metrics commonly used in LLM red teaming evaluations can broadly be grouped into four categories of assessing output quality, harm detection, attack effectiveness, and attack robustness.

In evaluating output quality, traditional metrics such as ROUGE (Lin, 2004) and BLEU (Papineni et al., 2002) remain foundational, focusing on measuring textual similarity between LLM-generated outputs and human-written references. These metrics help identify potential false positives in LLM adversarial scenarios (Lin, 2004; Papineni et al., 2002). Recently, BERTScore (Zhang* et al., 2020) has emerged as a promising metric that leverages contextual embeddings to evaluate text generation quality, offering a more nuanced assessment than traditional n-gram-based methods. For language modeling tasks, Perplexity (Jelinek et al., 1977) remains a fundamental metric, quantifying how well a model predicts a sample of text. Regarding text generation quality, MAUVE (Pillutla et al., 2021) offers a novel approach by comparing the distributions of human and machine-generated text.

For potential harm detection, metrics such as ToxicBLERT (Zhou et al., 2021) have been developed to identify and quantify harmful content generation. ToxicBLERT employs transformer-based classifiers to score text outputs for toxicity, enabling automated detection and filtering of toxic language from model outputs.

The assessment of attack effectiveness prominently features the ASR which is widely adopted for quantifying how often adversarial inputs successfully elicit harmful responses from LLMs (Cui, Zhang et al., 2024; Zhang et al., 2024). ASR can be evaluated through manual examination, rule-based keyword detection (Zou et al., 2023), or by leveraging other LLMs like GPT-4 (Achiam et al., 2023) to perform evaluations. Parametrized binary toxicity classifiers (He et al., 2023; Perez et al., 2023) can also be used to determine attack success (Gehman et al., 2020).

Metrics for attack robustness evaluate how stable or fragile adversarial strategies are to minor perturbations, reflecting the resilience of attacks themselves. For example, robustness metrics from Qiu et al. (2023) measure the generalizability and reliability of identified vulnerabilities by examining the sensitivity of successful adversarial attacks to slight input variations. On the other hand, metrics proposed by Wang et al. (2023a) and Yang, Meng et al. (2024) directly quantify LLM resistance to adversarial perturbations and unexpected inputs. Efficiency is another important consideration in attack evaluation, although standardized quantitative methods for measuring attack efficiency are still lacking (Chao et al., 2024; Zou et al., 2023).

A recent metric CC-SHAP (Parcalabescu & Frank, 2024) aims to measure the self-consistency-cum-faithfulness of natural language explanations generated by LLMs. Unlike traditional self-consistency tests that require semantic evaluations or input edits, CC-SHAP analyzes the contribution of each input token to both the model's answer prediction and the generated explanation. This consistency measure, based on SHAPley values, provides a continuous self-consistency value per instance and is interpretable, identifying individual token contributions and highlighting differences in input usage between prediction and explanation.

Beyond metrics, structured benchmarks are essential for standardized evaluation of LLM vulnerabilities across multiple adversarial contexts. To effectively simulate and evaluate red teaming scenarios from Section 4, selected benchmarks demonstrate key criteria of scalability, generalizability, and adaptability. Where, scalability enables consistent evaluation across diverse models and dataset sizes, generalizability ensures coverage of multiple attack vectors and real-world scenarios, and adaptability allows continuous integration of emerging adversarial methods. Table 8 presents an overview of benchmarks that meet these criteria by focusing on red teaming contexts. We observe a trend toward more diverse and challenging evaluation suites regarding the comprehensive

Table 6
Comprehensive overview of LLM evaluation datasets.

Reference	Name	Year	Samples	Structure	Labels	Collection	Annotation	Use Cases	Features	Domain
Hartvigsen et al. (2022)	ToxiGen	2022	250,000	Text passages	Toxicity, target group	AI-generated	Human verification	Toxicity detection, bias evaluation	Adversarially generated toxic content	Social media, online forums
Gehman et al. (2020)	RealToxicity Prompts	2020	100,000	Sentence prompts	Toxicity scores	Web scraping	Automated toxicity scoring	Prompt-based toxicity evaluation	Real-world toxic prompts	Web content
Wang et al. (2022)	AdvGLUE	2021	Varies by task	Multiple NLP tasks	Task-specific	Adversarial generation	Human verification	Robustness testing across NLP tasks	Adversarial examples for GLUE benchmark	Various NLP tasks
Huang, Zhang, Y et al. (2023)	TRUSTGPT	2023	26,000	Multi-turn dialogues	Safety violations, task completion	AI-generated dialogues	Human annotation	Evaluating LLM safety in dialogues	Multi-turn safety evaluation	Open-domain dialogues
Zou et al. (2023)	AdvBench	2023	5000	Text pairs	Attack type, success rate	Automated adversarial generation	Automated evaluation	Benchmarking robustness against attacks	Diverse attack types	General text
Zhang et al. (2024)	SafetyBench	2023	Over 10,000	Prompts and responses	Safety categories	Curated and generated	Expert annotation	Comprehensive safety evaluation	Multi-dimensional safety assessment	Various safety-critical scenarios
Zhao et al. (2018)	WinoBias	2018	3160	Sentences with pronoun resolution	Correct pronoun resolutions, occupation types	Manually created	Expert creation and validation	Evaluating gender bias	Balanced gender-occupation pairs	Everyday scenarios
Nangia et al. (2020)	CrowS-Pairs	2020	1508	Sentence pairs	Stereotypical vs. anti-stereotypical	Crowdsourced	Multi-stage human validation	Measuring social biases	Covers nine types of biases	General text, social scenarios
Lin et al. (2022)	TruthfulQA	2021	817	Question-answer pairs	Truthful vs. false answers	Manually curated	Expert creation and peer review	Evaluating truthfulness	Questions eliciting common misconceptions	General knowledge, current events
Röttger et al. (2021)	HateCheck	2021	3728	Short text snippets	Hate speech categories	Manually created	Expert creation and validation	Evaluating hate speech detection	29 functional tests for hate speech	Social media, online comments
Parrish et al. (2022)	BBQ	2022	58,492	Question-answer pairs	Demographic categories, correct answers	Template-based generation	Automated with human verification	Measuring demographic biases in QA	Covers 11 demographic categories	General knowledge, social scenarios
Nadeem et al. (2021)	StereoSet	2021	17,000	Context sentences with completions	Stereotype and anti-stereotype scores	Crowdsourced	Multi-annotator agreement	Measuring stereotypical biases	Covers gender, profession, race, religion	General text, social scenarios
Dhamala et al. (2021)	BOLD	2021	23,679	Prompts for text generation	Demographic attributes	Manually curated	Expert creation and validation	Evaluating bias in generation	Focuses on intersectional bias	Various contexts
Ganguli et al. (2022)	RedTeaming	2022	6800	Adversarial prompts	Harm categories	Expert-created	Expert evaluation	Stress-testing LLMs for safety	Diverse harm categories	Various (explicit content, misinformation)
Zellers et al. (2019)	RealNews	2019	3.4 million	News articles	Real vs. machine-generated	Web scraping and AI generation	Automated labeling	Detecting AI-generated fake news	Large-scale machine-generated text	News articles
Ji et al. (2023)	BEAVER-TAILS	2023	5000	Prompts and responses	Tailored instructions, safety violations	AI-generated with human curation	Expert evaluation	Evaluating instruction-following and safety	Focuses on bypassing safety measures	Various harmful scenarios
Bhardwaj and Poria (2023a)	Unalignment Dataset	2023	Not specified	Prompts for unaligned behavior	Unalignment categories	Expert-created	Expert evaluation	Identifying unaligned behavior in LLMs	Covers various aspects of AI alignment	Ethics, safety, alignment scenarios
Bhardwaj and Poria (2023b)	HARMFUL-QA	2023	Not specified	Question-answer pairs	Harm categories	Expert-created	Expert evaluation	Evaluating propensity to generate harmful content	Focuses on eliciting harmful information	Various harmful scenarios
Bhardwaj and Poria (2023b)	DANGEROUS-QA	2023	Not specified	Question-answer pairs	Danger categories	Expert-created	Expert evaluation	Assessing potential to provide dangerous information	Emphasis on physically dangerous scenarios	Potentially dangerous activities
Akun (2025)	Awesome ChatGPT Prompts	2024	176	Prompts for various tasks	Task categories	Community-contributed	Community curation	Evaluating LLM capabilities across tasks	Wide range of creative and practical prompts	Various (writing, coding, problem-solving)

(continued on next page)

Table 6 (continued).

Taori et al. (2023)	Alpaca	2023	52,000	Instruction-output pairs	Task categories	Self-instruct with GPT-3	AI-generated with human filtering	Fine-tuning for instruction-following	Diverse set of instructions and tasks	General tasks
Zhu et al. (2015)	BookCorpus	2015	11,038 books	Full-text books	None (raw text)	Web scraping	None (raw data)	Pre-training language models	Large collection of coherent, long-form text	Fiction books
Bai et al. (2022a)	Harmless Anthropic	2022	Not specified	Prompts and responses	Safety categories	AI-generated with human curation	Expert evaluation	Training and evaluating safe AI systems	Focus on capability with safety	Various (knowledge, tasks, ethics)
Zhu, Wang et al. (2023)	PromptBench	2023	583,884	Adversarial examples	Attack types	AI-generated	Automated	Evaluating LLM robustness	Wide range of text-based attacks	Various NLP tasks
Nie et al. (2020)	ANLI	2020	170,000	Sentence pairs	Entailment labels	Human-crafted adversarial examples	Expert annotation	Evaluating natural language inference	Adversarial examples with spelling errors and synonyms	Natural language inference
Yang, Zhang et al. (2023)	GLUE-X	2022	Not specified	OOD samples for GLUE tasks	Task-specific	Curated from various sources	Expert curation	Evaluating OOD generalization	14 groups of OOD samples	8 classic NLP tasks
Yuan et al. (2023)	BOSS	2022	Not specified	OOD samples for NLP tasks	Task-specific	Curated and generated	Expert curation	Evaluating OOD robustness	20 groups of samples for 5 NLP tasks	Various NLP tasks

Table 7
Metrics for LLM attack testing.

Reference	Name	Year	Description	Task/Use case	Range	Automated
Lin (2004)	ROUGE	2004	Evaluates quality of summaries	Text summarization, Attack false positive detection	0–1	Yes
Papineni et al. (2002)	BLEU	2002	Measures quality of machine translations	Machine translation, Attack false positive detection	0–1	Yes
Zhang* et al. (2020)	BERTScore	2019	Evaluates text generation quality using BERT embeddings	Text generation	–1 to 1	Yes
Jelinek et al. (1977)	Perplexity	Various	Measures how well a model predicts a sample	Language modeling	≥ 1	Yes
Pillutla et al. (2021)	MAUVE	2021	Compares distributions of human and machine-generated text	Text generation quality	0–1	Yes
Zhou et al. (2021)	ToxicBLERT	2021	Measures toxicity in language model outputs	Toxicity detection	0–1	Yes
Cui, Zhang et al. (2024) and Zhang et al. (2024)	ASR	2023	Measures success rate of eliciting harmful content	Attack effectiveness	0–1	Partially
Qiu et al. (2023)	Attack Robustness	2023	Assesses sensitivity of attacks to perturbations	Attack robustness	Varies	Partially
Parcalabescu and Frank (2024)	CC-SHAP	2024	Measures input-level self-consistency using SHAP values	Faithfulness and self-consistency	Continuous	Yes
Yang, Meng et al. (2024)	Adversarial Robustness	2024	Assesses LLMs resistance to adversarial perturbations	LLM sensitivity and robustness to attack	Varies	Partially
Wang et al. (2023a)	ChatGPT's robustness analysis	2023	Performance on unexpected inputs	adversarial and out-of-distribution robustness	Varies	Partially
Hoffmann et al. (2022)	Computational Efficiency	2022	Leading resource for compute optimal LLM training	Optimal parameters, training tokens, model scaling	Varies	Partially
Choudhury and Deshpande (2021)	Rawlsian fairness	2021	Assesses performance consistency of LLMs across multiple languages and tasks	Utilitarian multilingual model selection	Varies	Partially

benchmarks. The General Language Understanding Evaluation (GLUE) benchmark ([Wang et al., 2018](#)) and its more difficult successor SuperGLUE ([Wang et al., 2019](#)) have become standard tools for assessing natural language understanding across a range of tasks. These have inspired more specialized benchmarks like AdvGLUE ([Wang et al., 2021](#)), which focuses on adversarial examples to test model robustness.

For evaluating LLMs resistance to prompt-based attacks, benchmarks such as JailTrickBench ([Xu et al., 2024](#)), and Adversarial Prompt Robustness ([Wang et al., 2023b](#)) provide systematic assessment frameworks designed for prompt injection and jailbreaking scenarios. These benchmarks utilize diverse adversarial prompts to test safety training effectiveness and robustness to manipulation, which is critical for identifying vulnerabilities in prompt-aligned safety mechanisms.

To assess model vulnerabilities arising from data manipulation and knowledge poisoning, CTIBench ([Alam et al., 2024](#)) and CyberSecEval 2 ([Bhatt et al., 2024](#)) benchmarks simulate realistic cyber threat intelligence and injection attacks scenarios. These structured tests evaluate model performance in the face of adversarial data inputs and compromised retrieval systems, capturing models' resilience against integrity degradation attacks.

Comprehensive benchmarks like AdvGLUE ([Wang et al., 2021](#)) and Adversarial NLI ([Nie et al., 2020](#)) provide extensive adversarial testing across broader linguistic tasks, emphasizing model exploitation and evasion techniques. These benchmarks challenge LLMs with carefully crafted adversarial examples across standard natural language understanding tasks, assessing the general robustness of model architectures and prompting mechanisms under adversarial pressures.

Table 8
Benchmarks for LLM attack testing and evaluation.

Ref	Name	Year	Category	Description	Task/Use case	Range	Auto-mated
Zhao et al. (2020)	Bias Benchmarks	2020	Fairness/Bias Evaluation	Assesses various types of bias in LLM outputs	Fairness evaluation	Varies	Partially
Sakaguchi et al. (2021)	WinoGrande	2020	Common-sense Reasoning	Large-scale dataset for commonsense reasoning	Commonsense reasoning	Accuracy (%)	Yes
Gehman et al. (2020)	RealToxicity Prompts	2021	Safety Evaluation	Dataset of prompts likely to elicit toxic responses	Toxicity evaluation	N/A	Requires human
Lin et al. (2022)	TruthfulQA	2021	Truthfulness Evaluation	Measures model's tendency to generate false or misleading information	Truthfulness assessment	0–1	Partially
Nadeem et al. (2021)	StereoSet	2021	Bias Detection	Measures stereotype bias in language models	Bias detection	0–100	Yes
Chen et al. (2021)	HumanEval	2021	Code Generation Evaluation	Evaluates code generation capabilities	Programming tasks	Pass@k (%)	Yes
Bai et al. (2022b)	Anthropic's Constitutional AI Benchmark	2022	Ethical/Behavior Evaluation	Evaluates model's adherence to specified behavioral constraints	Ethical behavior assessment	Varies	Partially
Huang, Zhang, Sun et al. (2023)	TrustGPT	2023	Trustworthiness Evaluation	Evaluates trustworthiness of LLM outputs	Trustworthiness assessment	0–1	Partially
Miller (2024)	Anthropic Evals	2023	Multi-faceted Evaluation	Suite of evaluations for model capabilities and safety	Multi-faceted evaluation	Varies by task	Partially
Cui, Chiang et al. (2024)	Over-Refusal Rate	2024	Safety Override	Measures the frequency at which the model incorrectly refuses to respond under adversarial or safety-triggering inputs	Safety evaluation	0–1	Partially
Nie et al. (2020)	Adversarial NLI	2020	Adversarial/Red Teaming	Tests model robustness to adversarial examples	Natural language inference	Accuracy (%)	Yes
Wang et al. (2021)	AdvGLUE	2021	Adversarial/Red Teaming	Adversarial version of GLUE benchmark	Natural language understanding	Varies by task	Yes
Wang et al. (2023b)	Adversarial Prompt Robustness	2023	Adversarial/Red Teaming	Measures performance under adversarial prompts	Robustness evaluation	Varies	Partially
Bhatt et al. (2024)	CyberSecEval 2	2024	Adversarial/Red Teaming	Evaluates LLM security risks including prompt injection and code interpreter abuse	Cybersecurity evaluation	Varies by task	Partially
Alam et al. (2024)	CTIBench	2024	Adversarial/Red Teaming	Benchmark suite for evaluating LLM performance in cyber threat intelligence	Cyber threat intelligence evaluation	Varies by task	Partially
Xu et al. (2024)	JailTrick-Bench	2024	Adversarial/Red Teaming	Benchmark for assessing diverse jailbreak attacks on LLMs	Jailbreak attack evaluation	Varies	Yes

Several additional benchmarks focus on targeted ethical and safety concerns crucial to red teaming evaluations. TruthfulQA (Lin et al., 2022) assesses LLMs tendency to generate misleading or false information, highlighting vulnerabilities relevant to misinformation and harmful content generation attacks. Similarly, StereoSet (Nadeem et al., 2021) evaluates stereotype biases as indirect consequences of adversarially induced biases or data poisoning attacks.

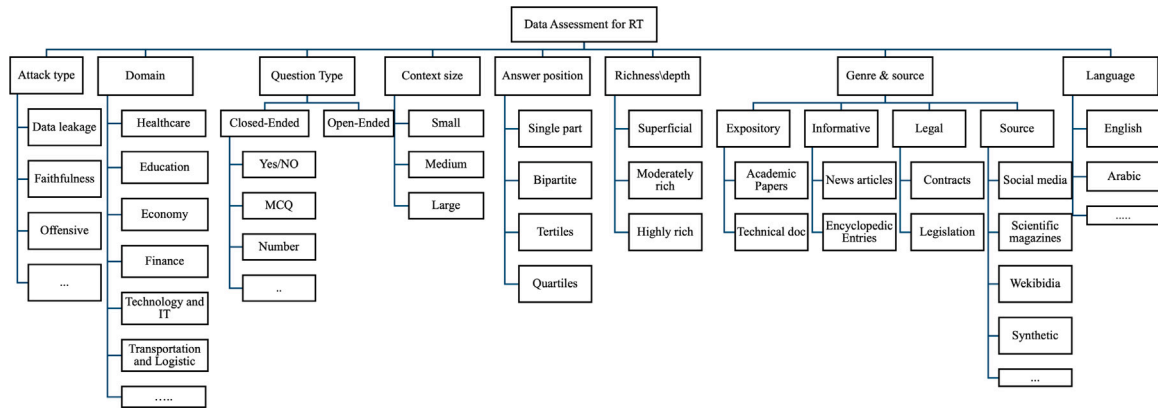


Fig. 5. Assessment framework for data utilized in LLM red teaming.

For more holistic and multi-dimensional evaluation approach, Anthropic Evals (Miller, 2024) and Anthropic's Constitutional AI Benchmark (Bai et al., 2022b) incorporate various evaluation dimensions, from ethical adherence to robustness against red teaming scenarios, providing structured frameworks to assess overall safety and alignment of LLM outputs comprehensively. Additionally, the Over-Refusal Rate benchmark (Cui, Chiang et al., 2024) evaluates models' safety overrides, measuring how often models incorrectly refuse valid prompts under adversarially induced safety triggers, thus identifying overprotective misalignments resulting from safety training mechanisms. TrustGPT (Huang, Zhang, Sun et al., 2023) aims to assess the trustworthiness of LLM outputs. HELM (Liang et al., 2023) offers a holistic approach to LLM evaluation, while EleutherAI's Language Model Evaluation Harness (Gao et al., 2021) provides a suite of benchmarks for various LLM capabilities. The BIG-bench (Srivastava et al., 2023) goes beyond traditional tasks to include a diverse set of challenges that push the boundaries of what we expect from language models. It is worth noting that many of these benchmarks, such as RealToxicityPrompts (Gehman et al., 2020) and Anthropic's Constitutional AI Benchmark (Bai et al., 2022b), require some degree of human evaluation, highlighting the ongoing importance of human judgment in assessing LLM performance and safety.

These metrics and benchmarks collectively enhance our capability to rigorously evaluate LLM safety, robustness, and resilience to diverse red teaming attacks.

5.3. Data evaluation and benchmarking

We propose a data-taxonomy for data generation for red teaming illustrated in Fig. 5. It considers several aspects including:

- Domain: Broad data domains include healthcare, education, finance, technology, and more.
 - Question Type: Questions may be open-ended or close-ended. Close-ended types include Boolean (Yes/No), multiple choice, numerical, or single-word answers.
 - Data Source: Origin or sources include websites, repository, social media, research databases, scientific magazines, and news outlets.
 - Context Size: Contexts are classified as small (fewer than 1024 tokens), medium (1024–3072 tokens), or large (more than 3072 tokens).
 - Answer Position:
 - Single part: Not considering the answer position, whole document considered as one part.
 - Bipartite: Two parts, with answer in either part.
 - Tertiles: Once considering the context as three parts.
 - Quartiles: Once considering the context as four parts.
 - Richness & Depth:
 - Superficial: Minimal context depth, leading to fewer questions or insights.
 - Moderately rich: Moderate depth, moderate questions.
 - Highly rich: Significant depth, many questions.
- Ratio of text size to the number of questions determines the classification.
- Genre: Data can be categorized based on its purpose, style, and structure etc.
 - Narrative: Fiction (novels, short stories) or non-fiction (biographies, memoirs).

- Expository: Academic papers, technical documentation.
 - Descriptive: Product descriptions, travel writing.
 - Persuasive: Editorials, advertising texts.
 - Informative: News articles, encyclopedic entries.
 - Dialogue: Scripts, transcriptions.
 - Poetic: Poetry.
 - Legal: Contracts, legislation.
 - Social Media: Tweets, status updates.
- Language: Languages based on data context.

We have evaluated the red teaming datasets using the proposed framework, and the key characteristics are presented in [Table 9](#). The primary purpose of the data-taxonomy and its application in [Table 9](#) is to provide a structured categorization of existing datasets and to aid datasets development rather than to present empirical analyses or frequency-based statistical insights.

It is clear that a variety of tasks have been taken into account in the development of datasets for evaluation purposes. Most of these datasets have been designed for general domains. However, nearly all of the evaluated studies have focused on open-ended questions, often neglecting considerations such as context size and answer positioning. Additionally, there is a noticeable variation in the genre of the data. The datasets in question may include synthetic data, manually generated content, or data evaluated by domain experts. Furthermore, all the datasets reviewed are exclusively in English.

While the range of datasets used for LLM red teaming is extensive and diverse, they inherently carry biases that may inadvertently reinforce harmful patterns in language models. Many of these datasets are predominantly derived from web-scraped content, social media, and other publicly available sources, which reflect the prevailing societal biases and toxic language present in those domains. Models trained or evaluated on such data can internalize these biases as a result, leading to outputs that perpetuate stereotypes, misinformation, or other forms of harmful content. Moreover, evaluation datasets that rely on human annotation or crowdsourcing may introduce subjectivity and cultural bias, particularly when annotators come from a limited range of cultural or linguistic backgrounds. This can lead to over-representation of dominant cultural norms and an underrepresentation of diverse perspectives.

From a technical standpoint, biases in these datasets affect the statistical distribution of training examples which causing models to attend to patterns that are common in the source data, even if those patterns are harmful. When these biased datasets are used in red teaming evaluations, they may not only fail to penalize such behaviors but might also inadvertently validate them, as metrics such as ASR and toxicity scores can be skewed by the underlying data distribution. In adversarial contexts, this issue is compounded if the evaluation frameworks do not account for such biases, potentially masking the true severity of vulnerabilities or overestimating a model's robustness. Therefore, it is crucial that future research aims to develop more balanced, multilingual, and culturally diverse datasets, along with evaluation metrics that can detect and adjust for these biases. Such advancements are essential for ensuring that LLMs are assessed in a manner that reflects their potential impact on a global and heterogeneous user base.

This framework sets the foundation for future empirical studies where detailed analyses can be conducted to quantitatively examine these datasets, explore statistical distributions, identify biases, and understand the impacts of specific dataset attributes on LLM robustness, vulnerabilities, or security concerns.

6. Analysis and discussion

Throughout this review, we have thoroughly addressed the research questions that guided our inquiry ([Section 2.1](#)). We have identified major trends in LLM red teaming, categorized various methodologies, and explored the primary tasks these efforts aim to address. Our discussion of LLM vulnerabilities and exploits, along with an analysis of key failure modes provides a comprehensive understanding of the threats faced by these systems. We have also examined strategies for developing effective red teaming exercises and highlighted the most significant approaches for assessing LLM vulnerabilities and attacks.

To synthesize our findings, the next two subsections respectively revisit the primary research questions and discuss current limitations in LLM red teaming along with directions for future research.

6.1. Reflections on the research questions

In this subsection, we revisit the research questions outlined at the outset, drawing together insights from the reviewed literature to highlight key findings and emerging themes.

RQ1: What is the current state of research in LLM red teaming? One of the prominent trends in examining the LLM red teaming landscape is the increasing sophistication and variety in attack methodologies. The adversaries are leveraging more advanced techniques as LLMs become more capable, such as prompt-based attacks and data manipulation to bypass safety mechanisms and induce harmful behaviors. Please refer to [Table 10](#) covering strengths and weaknesses of significant red teaming studies. Another significant trend is the shift toward more automated and scalable red teaming processes, exemplified by the development of tools like AutoDAN and CIA, which automate the generation of harmful prompts.

RQ2: What taxonomy could be used to represent and classify LLM red teaming attacks? Red teaming for LLMs encompasses a diverse array of methodologies, each tailored to exploit specific vulnerabilities. Our review has classified these methodologies into five primary categories of prompt-based attacks, data manipulation, model exploitation, information extraction, and model degradation ([Fig. 3](#)).

Table 9
Datasets evaluation using the proposed framework.

Reference	Task	Domain	Query type	Context size	Answer position	Richness	Genre	Source	Language	LLM name	Attack type	Approach	RAG or LLMs
Hartvigsen et al. (2022)	Toxicity detection	Social	Question/ Answer	NA	NA	High	Social media	AI-generated	English	NA	Adversarial	Human verification	LLMs only
Gehman et al. (2020)	Prompt-based eval	General	Open-ended	NA	NA	Medium	Web content	Web scraping	English	NA	NA	Automated scoring	LLMs only
Wang et al. (2022)	NLP tasks	General	Open-ended	Varies	Task-Specific	High	Various NLP	Adversarial generation	English	NA	Adversarial	Human verification	LLMs only
Huang, Zhang, Y et al. (2023)	Safety evaluation	General	Open-ended	NA	NA	High	Open-domain	AI-generated	English	NA	NA	Human annotation	LLMs only
Zou et al. (2023)	Robustness testing	General	Open-ended	NA	NA	High	General text	AI-generated	English	NA	Adversarial	Automated	LLMs only
Zhang et al. (2024)	Safety assessment	General	Open-ended	NA	NA	High	Safety-critical	Curated and generated	English	NA	NA	Expert annotation	LLMs only
Zhao et al. (2018)	Bias evaluation	General	Question/ Answer	NA	NA	Medium	Everyday scenarios	Manually created	English	NA	NA	Expert validation	LLMs only
Nangia et al. (2020)	Bias measurement	General	Open-ended	NA	NA	Medium	General text	Crowdsourced	English	NA	NA	Human validation	LLMs only
Lin et al. (2022)	Truthfulness eval	General	Question/ Answer	NA	NA	Medium	General knowledge	Manually curated	English	NA	NA	Expert review	LLMs only
Röttger et al. (2021)	Hate speech detection	Social	Question/ Answer	NA	NA	High	Social media	Manually created	English	NA	NA	Expert validation	LLMs only
Parrish et al. (2022)	Bias measurement	General	Open-ended	NA	NA	High	General knowledge	Template-based	English	NA	NA	Human verification	LLMs only
Nadeem et al. (2021)	Bias measurement	General	Open-ended	NA	NA	High	General text	Crowdsourced	English	NA	NA	Multi-annotator agreement	LLMs only
Dhamala et al. (2021)	Bias evaluation	General	Open-ended	NA	NA	High	Various contexts	Manually curated	English	NA	NA	Expert validation	LLMs only
Ganguli et al. (2022)	Safety stress-testing	General	Open-ended	NA	NA	High	Various	Expert-created	English	NA	NA	Expert evaluation	LLMs only
Zellers et al. (2019)	Fake news detection	News	Open-ended	NA	NA	High	News articles	Web scraping & gen-AI	English	NA	NA	Automated labeling	LLMs only
Ji et al. (2023)	Instruction-following and safety	General	Open-ended	NA	NA	High	Various	AI-generated with human curation	English	NA	NA	Expert evaluation	LLMs only
Bhardwaj and Poria (2023a)	AI alignment detection	General	Open-ended	NA	NA	High	Ethics, safety	Expert-created	English	NA	NA	Expert evaluation	LLMs only
Bhardwaj and Poria (2023b)	Harm detection	General	Question/ Answer	NA	NA	High	Various	Expert-created	English	NA	NA	Expert evaluation	LLMs only
Bhardwaj and Poria (2023b)	Dangerous info detection	General	Question/ Answer	NA	NA	High	Potentially dangerous activities	Expert-created	English	NA	NA	Expert evaluation	LLMs only
Akn (2025)	LLM capabilities eval	General	Open-ended	NA	NA	Medium	Various	Community-contributed	English	NA	NA	Community curation	LLMs only
Taori et al. (2023)	Instruction-following	General	Open-ended	NA	NA	Medium	General tasks	AI-generated with human filtering	English	GPT-3	NA	Self-instruct	LLMs only
Zhu et al. (2015)	Language model pre-training	Fiction	Open-ended	NA	NA	Medium	Fiction books	Web scraping	English	NA	NA	None	LLMs only
Bai et al. (2022a)	Safe AI training	General	Open-ended	NA	NA	High	Various	AI-generated with human curation	English	NA	NA	Expert evaluation	LLMs only
Zhu, Wang et al. (2023)	LLM robustness evaluation	General	Open-ended	NA	NA	High	Various NLP tasks	AI-generated	English	NA	Adversarial	Automated	LLMs only

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Table 9 (continued).

Nie et al. (2020)	NLI evaluation	General	Open-ended	NA	NA	Medium	Natural language inference	Human-crafted adversarial examples	English	NA	Adversarial	Expert annotation	LLMs only
Yang, Zhang et al. (2023)	OOD generalization	General	Open-ended	NA	NA	Medium	NLP tasks	Curated from various sources	English	NA	NA	Expert curation	LLMs only
Yuan et al. (2023)	OOD robustness	General	Open-ended	NA	NA	Medium	NLP tasks	Curated and generated	English	NA	NA	Expert curation	LLMs only

Table 10

Strengths and limitations of key studies on LLM red team attacks.

Reference	Strengths	Limitations
Wei et al. (2023)	Good insights from safety training failure cases.	Attack type not generalized; focus only on jailbreak attacks.
Tong et al. (2023)	Mass-production of red team test cases in text-guided multimodal systems; Novel direction.	Limited to text-guided multimodal systems; qualitative assessments only.
Wichers et al. (2024)	Gradient-based automated red teaming prompt discovery, high success rate.	Limited red teaming attack choice; token limits for prompt and response.
Mazeika et al. (2024)	Automated red teaming, Open-source classifier	
Naito et al. (2023)	Automated penetration testing	Small number of devices tested; technique relies on known exploits.
Zou et al. (2023)	Universal multi-model attacks; works on open-source and black-box LLMs.	
Pelrine et al. (2024)	Demonstrated API-based vulnerabilities such as safety guardrails removal, private info leaking, injection attacks.	API exploits may not be generalizable for broader red teaming mitigation.
Hong et al. (2024)	Greater coverage of toxic test cases; thorough evaluation and analysis.	Attack performance depends on red teaming LLM parameters such as novelty reward weight.
Geiping et al. (2024)	Multiple attack examples for unintended behaviors in models.	Scalability to open-source and black-box models unclear.
Singh et al. (2023)	Persuasive conversations analyzed for vulnerabilities.	Hard to scale; limited test cases; lacks insights on affirmative responses and mitigation.
Ma et al. (2024)	Automated diverse attack strategies aiding empirical improvement in security of models.	Limited evaluation scope affects generalizability.
Qiang et al. (2024)	Novel GGI algorithm for adversarial attacks; comprehensive evaluation of transferability and model scale impacts.	Defense methods not suggested.
Chao et al. (2024)	Efficient jailbreak generation with fewer than 20 queries; good transferability.	Focus primarily on efficiency, does not cover robustness across attack dimensions.
Zheng et al. (2024)	High ASR across models (>95%); effective against advanced defenses.	Potentially less effective against token obfuscation; applicability to closed-source LLMs uncertain.
Guo et al. (2024)	High controllability and diversity of attacks. Good transferability.	Focuses primarily on white-box attacks, limiting applicability to black-box models.
Jia et al. (2024)	100% success rates on various models. Good transferability to both open and closed-source LLMs.	Defense methods not suggested.
Qi et al. (2024)	Effectiveness of visual adversarial examples. Generalizes beyond optimization corpus.	Fundamental challenges in protecting against such attacks, especially for offline models.
Huang, Wang et al. (2024)	Incorporation of prompt semantics for improved attack effectiveness; tested across multiple models and response types.	Focused on semantic aspects, may need more exploration of other dimensions.
Wang, Li, Huang et al. (2024)	Reduced computation time for generating adversarial suffixes. Improved textual fluency. Generalizable to multiple LLMs.	Potential challenges in adapting to all types of adversarial suffixes. May require specific training for different LLMs.
Qiang et al. (2024)	High ASR and task manipulation effectiveness. Transferable across models and datasets. Effective with more in-context examples.	Defense methods not suggested.
Zhu, Zhang et al. (2023)	Diverse and readable prompts. Effective at evading perplexity filters. Generalizes to new harmful behaviors. Versatile with black-box models.	May require adaptation to different LLMs. Performance can be affected by model-specific defenses.
Zou et al. (2024)	High ASR with minimal poisoned texts. Applicable to large databases. Demonstrates insufficiency of current defenses.	Defense methods not suggested.
Cui, Chiang et al. (2024)	Comprehensive evaluation.	Focuses on measuring over-refusal without direct attack strategy.
Jeong (2024)	Extends adversarial attack research to multi-modal domains.	Limited quantitative evaluation. Mitigation struggles when irrelevant contexts dominate.
Wang, Li, Jiang et al. (2024)	Efficient and robust method for testing tokenization.	Focuses on tokenization issues without addressing specific defensive measures. Applicable to open-source LLMs only.

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Table 10 (continued).

Perez and Ribeiro (2022)	Provided a systematic framework for testing adversarial prompts. Highlighted inverse scaling phenomenon in LLMs.	Requires stop sequences and post-processing for effective mitigation. Key vulnerabilities identification in GPT-3 only.
Greshake et al. (2023)	Novel attack vector for LLM-integrated applications. Comprehensive taxonomy for new security risks.	Lack of effective mitigations for emerging threats.
Melamed et al. (2024)	Transferability and robustness of evil twin prompts. Applications in prompt compression and conditional generation.	Challenges in maintaining functional similarity across different token replacements and LLM families.
Liu, Deng, Li et al. (2024)	Effectiveness for LLM-integrated applications. Severe attack outcomes demonstration; such as arbitrary LLM usage and application prompt theft.	Defense strategies like instruction defense, post-prompting, and others found insufficient.
Li, Zhou et al. (2024)	High success rates. Effective with both open-source and closed-source models. Method is lightweight and avoids brute-force optimization.	Relies on the model's imagination capability and personification, which may vary across different models.
Ding et al. (2024)	Automated prompt generation without additional training or optimization. High ASR. Effectiveness in bypassing current defenses.	Relies on the model's ability to follow nested scenarios, which may vary across different models.
Yong et al. (2024)	Linguistic inequality vulnerability analysis in LLM safety training. Effectiveness with simple translations.	Focuses on low-resource languages which may not cover all potential attack scenarios.
Mehrotra et al. (2024)	Systematic refinement of prompts. Efficient with fewer queries. Outperforms previous black-box methods.	Requires coordination between multiple LLMs. May be complex to implement without a clear understanding of the reasoning process.
Shah, Feuillade-Montixi et al. (2023)	Automated technique. High harmfulness completion rates.	Fully automated approach may not generate the most harmful outputs. Requires human oversight for optimal results.
Shah, Sharma et al. (2023)	Employs local fine-tuning to reduce divergence between proxy and target models. Good attack transferability.	Success does not always guarantee harmful content. Metrics may not accurately measure attack success.
Shu et al. (2023)	Alters model behavior with minimal data poisoning. High stealthiness due to natural integration with benign data.	Potential challenges in generalizing the attack across different LLM architectures.
Vega et al. (2024)	Simple and optimization-free attack method. Improved ASR. Automated evaluation pipeline.	Dependent on manual crafting of initial responses. No rigorous human study was conducted for attack crafting.
Chen, Tang et al. (2024)	Amplifies privacy risks 10x. Effective against commercial LLMs' APIs. Overcomes hallucination and catastrophic forgetting.	Dependent on fine-tuning interface availability. Real-world impact on deployed systems.
Lukas et al. (2023)	Empirical evidence of DP's limitations in preventing PII exposure. Novel attack strategies.	Scrubbing reduces utility. DP leaks 3% of PII. 3. Dataset curation techniques insufficient for complete protection.
Qi et al. (2023)	LLM Vulnerability analysis to fine-tuning. Safety degradation even with benign datasets. Cost-effective attack implementation.	Limited to current models and datasets.
Zhan et al. (2024)	Effective bypass of RLHF protections. High success rate with minimal examples.	Limited to current fine-tuning techniques and models.
Jiang, Chen et al. (2023)	High ASR (95%+). Effective in bypassing safety mechanisms.	May not address all forms of complex malicious intents. Requires sophisticated prompt design.

Prompt-based attacks focus on crafting inputs that deceive the model into generating harmful content. Data manipulation involves altering training data to introduce vulnerabilities during model development. Model exploitation attacks seek to uncover and exploit inherent weaknesses in the model's architecture or training process. Information extraction aims to illicitly obtain sensitive or proprietary information from the model. Lastly, model degradation attacks are designed to reduce the overall performance or reliability of the model. These primary attack categories are further organized in sub-categories, covering a broad range of attacks. which provides a comprehensive framework for understanding the different approaches to red teaming and their respective targets within LLM systems.

RQ3: What are the major developments in the LLM teaming attacks? LLM red teaming efforts are primarily focused on identifying and mitigating risks related to the generation of harmful content, the leakage of sensitive information, and the exploitation of model vulnerabilities. Key tasks include stress-testing LLMs to uncover hidden weaknesses, simulating adversarial scenarios to assess the model's robustness, and developing countermeasures to prevent unsafe models deployment.

RQ4: How can effective LLM red teaming exercises be developed? Developing effective red teaming exercises for LLMs requires a multi-faceted approach that includes the careful selection of attack methodologies, the use of diverse datasets, and the continuous updating of threat models. Effective red teaming exercises should simulate realistic adversarial scenarios, incorporate a wide range of potential attack vectors, and evaluate the model's performance under different conditions. Additionally, these exercises should be iterative, with feedback loops that allow for the refinement of both the red teaming process and the model's defenses.

RQ5: What are the key failure modes that lead to risks and vulnerabilities in LLMs? LLM vulnerabilities often arise from specific failure modes, which can be broadly categorized into data-related, model-related, and deployment-related failures.

1. **Data-related Failures:** Data-related failures include issues such as biased or adversarial training data, which can lead to harmful outputs. The LLMs are trained on vast corpora that often include biased, noisy, or adversarial content. When the training data is skewed or contains misinformation, the resulting model may generate outputs that perpetuate harmful stereotypes or factual inaccuracies. For example, biased training data can lead to discriminatory responses in sensitive applications (Ashcroft & Whitaker, 2024). Moreover, the unintentional injection of adversarial examples during data collection can lead the model to learn spurious correlations. The resulting vulnerability might later be exploited by adversaries to induce harmful outputs, and such issues are particularly evident in domains where data quality and representativeness are variable.
2. **Model-related Failures:** Model-related failures arise from the inherent limitations of the model architecture or the inadequacies of the training process, which may leave the model susceptible to exploitation. A primary model related failure mode is competing objectives during training. As modern LLMs are typically optimized for multiple objectives; such as language modeling, instruction following, and safety alignment. This multifaceted optimization can lead to conflicts, which can result in an instruction-following bias. In this scenario, the model may produce harmful outputs to comply with user prompts, as observed in adversarial jailbreak techniques like the do anything now (DAN) attack (Shen et al., 2024), where the model is coerced into bypassing its safety mechanisms.

LLMs are pre-trained on diverse unfiltered datasets, and then undergo safety tuning on a more restricted set of examples. This divergence between the broad and unfiltered pretraining data, and the narrower scope of safety tuning can cause models to learn mismatched generalization between pretraining and safety training. This discrepancy can lead to cases where a model performs well in standard tasks but fails to apply robust safety measures when confronted with out-of-distribution or obfuscated inputs. Recent work has shown that models such as GPT-4 can produce highly accurate outputs under structured contexts (e.g., in RAG) yet remain vulnerable to hallucinations when their internal knowledge conflicts with external evidence (Xu, Qi et al., 2024). Research also shows that the parameters responsible for enforcing safety are often sparse within the vast model. And minor perturbations whether from pruning, low-rank modifications, or fine-tuning can disproportionately compromise these regions without significantly affecting overall performance (Dong, Mu et al., 2024). Overconfidence and poor calibration further aggravate risks, with models frequently generating overly certain responses even when the input is ambiguous (Pawitan & Holmes, 2024).

3. **Deployment-related Failures:** Even robustly trained LLMs can exhibit failure modes during deployment. Improper integration or configuration can expose the model to vulnerabilities that were not apparent under controlled testing conditions. For instance, inadequate runtime monitoring or misconfigurations may result in contextual mismatches, where a model that performs well on curated benchmarks fails when confronted with diverse real-world inputs. These deployment-related failures underscore the need for continuous monitoring and adaptive safety protocols to ensure that the model's behavior remains secure across varying operational environments (Jiangm, Xu et al., 2023; Liu, Deng, Meng et al., 2024).

It is important to note that failure modes are not uniformly distributed across all LLMs or application domains. High-stakes domains such as healthcare and financial services demand particularly rigorous safety measures, and even minor deviations in factual accuracy or fairness can have severe consequences in these areas. Conversely, models deployed in general-purpose applications may exhibit different vulnerability patterns, emphasizing the need for domain-specific evaluations. Recently, evaluations such as CyberSecEval 2 (Bhatt et al., 2024) and CTIBench (Alam et al., 2024) have revealed how failure modes manifest differently across domains, guiding the development of domain-specific mitigation strategies. The risks and vulnerabilities in LLMs are the result of complex interactions between data quality, model architecture, training objectives, and deployment practices. Addressing these challenges requires a versatile approach; incorporating robust data curation, careful tuning of competing objectives, and vigilant real-world monitoring. As the field evolves, continued research into these failure modes and the systematic evaluation frameworks remains critical for ensuring trustworthy LLMs.

RQ6: What datasets and metrics are used for LLM red team attacks? Common datasets contain a variety of adversarial examples, while metrics measure success rates, impact, and model behavior under stress. The evaluation of LLMs against red team attacks utilizes a variety of specialized datasets. Notable among these are ToxiGen and RealToxicityPrompts for toxicity detection, CrowS-Pairs and StereoSet for bias evaluation, and TRUSTGPT and SafetyBench for assessing safety and ethical considerations. Robustness is tested using datasets like AdvGLUE, AdvBench, and PromptBench. Additionally, datasets such as TruthfulQA are critical for evaluating the truthfulness and misinformation detection capabilities of LLMs. Also, several metrics are employed to measure the performance of LLMs against these datasets. Traditional metrics like ROUGE and BLEU are used to assess text generation quality, while more advanced metrics such as BERTScore and MAUVE offer a nuanced evaluation. Toxicity-specific metrics, including ToxicBLERT and ASR, are essential for understanding the potential harms in LLM outputs. These metrics collectively enable a comprehensive assessment of LLM vulnerabilities under red team attacks.

RQ7: What evaluation frameworks and measures are employed to assess LLM red teaming effectiveness? The assessment of red teaming efforts is conducted through a variety of evaluation frameworks and measures, including quantitative metrics such as success rates of attacks, qualitative assessments of model behavior, and the use of standardized benchmarks. Evaluation frameworks for LLM red teaming incorporate both automated pipelines and human-in-the-loop methods. Automated evaluation pipelines leverage adversarial attack libraries and testing suites to simulate and assess the model's performance under various threat scenarios. Comprehensive benchmarks such as GLUE, SuperGLUE, and AdvGLUE provide a robust framework for assessing natural language understanding

and robustness. Also, measures like ROUGE, BLEU, and BERTScore assess text quality, while metrics such as MAUVE compare distributions of human and machine-generated text. ASR is pivotal for measuring the effectiveness of adversarial attacks. Human-in-the-loop evaluations, involving expert reviews and qualitative insights, are crucial for capturing context-dependent vulnerabilities that automated methods may overlook. These combined approaches ensure a thorough and realistic assessment of LLM red teaming effectiveness.

RQ8: What are the major considerations for trustworthiness evaluation of LLMs, to ensure their reliable and ethical deployment in various applications? Trustworthiness plays a crucial role in ensuring the reliable and ethical use of LLMs in society. It is important to note that trustworthiness is a composite framework that encompasses multiple dimensions; including faithfulness, safety, robustness, fairness, and privacy. Unlike traditional machine learning models, LLMs possess sophisticated generative capabilities that can produce unpredictable outputs, introducing unique challenges for establishing trust (Huang, Sun et al., 2024; Ozkaya, 2023). This unpredictability raises significant concerns about potential harmful behaviors, including misinformation propagation and malicious misuse (Hadi et al., 2023; Sun et al., 2024). Achieving trustworthiness in LLMs involves several key considerations. Fig. 6 presents a hierarchical breakdown of the LLM trustworthiness framework, illustrating the main components and their subcomponents. At the foundation of trustworthiness is truthfulness, which ensures that LLMs produce accurate and factually correct information. Safety is another vital aspect, particularly in sensitive fields such as healthcare where inaccuracies can lead to significant consequences. Fairness is equally important, as it seeks to avoid bias and discrimination. Achieving fairness requires implementing strategies to mitigate potential biases present in the diverse datasets used for training these models. Moreover, robustness is essential for LLMs as they must maintain reliable performance across various conditions (especially in adversarial scenarios). Privacy also cannot be overlooked, thereby protecting personal data is vital as LLMs should respect user autonomy and dignity.

The trustworthiness framework for LLMs encompasses several critical dimensions that must be systematically addressed:

1. **Correctness and Truthfulness:** At the foundation of trustworthiness lies the model's ability to generate factually accurate information. Recent work by Maynez et al. (2020) introduces robust evaluation frameworks for measuring LLM truthfulness, highlighting the importance of knowledge attribution and verification. Furthermore, Xu, Qi et al. (2024) demonstrate that LLMs can exhibit significant hallucination tendencies, emphasizing the need for rigorous fact-checking mechanisms.
2. **Robustness and Reliability:** LLMs must maintain consistent performance across diverse inputs and operating conditions. Dong, Mu et al. (2024) identify that model robustness encompasses: adversarial robustness against malicious inputs, distribution shifts in deployment scenarios, and consistency in reasoning and output generation.
3. **Faithfulness:** This dimension ensures that LLM outputs accurately reflect the provided context and the logical reasoning applied during generation, without introducing unsupported or hallucinated details. Work by Lanham et al. (2023) demonstrates the importance of faithful reasoning chains in building trustworthy AI systems, particularly through techniques like chain-of-thought prompting and self-consistency verification. Recent research by Wu, Wu et al. (2024) reveals a critical tension between LLMs' internal knowledge and retrieved information in RAG systems, showing that while RAG can significantly improve accuracy (from 34.7% to 94% in GPT-4), effectiveness varies based on the model's prior confidence and prompt strategy. Addressing these challenges, Nguyen et al. (2024) introduces a specialized 9-billion parameter model for RAG applications alongside ContextualBench, a comprehensive evaluation framework. Their model, featuring an innovative chat template with "Thought" and "Observation" roles, demonstrates superior contextual faithfulness compared to larger models while maintaining strong general capabilities across multiple benchmarks.
4. **Safety Guarantees:** Particularly crucial in high-stakes domains like healthcare (Yang, Tan et al., 2023) and financial services (Li et al., 2023), where model outputs can have significant real-world consequences. Inan et al. (2023) propose comprehensive safety frameworks that include runtime monitoring systems, guardrails for harmful content, and uncertainty quantification methods.
5. **Fairness and Bias Mitigation:** Building on existing concerns about bias, recent work by Echterhoff et al. (2024) and Eigner and Händler (2024) introduces sophisticated approaches for identifying and measuring various forms of bias in LLM outputs, implementing debiasing techniques during training and inference, and ensuring equitable performance across different demographic groups.

These dimensions are deeply interconnected, as demonstrated by comprehensive studies (Huang, Sun et al., 2024) showing how improvements in one aspect often influence others. For instance, enhancing model robustness frequently leads to improved faithfulness (Li et al., 2025) and reliability in real-world applications. Achieving trustworthiness requires the systematic evaluation of all these dimensions.

A particularly critical consideration often overlooked in this landscape is faithfulness, which refers to the requirement that an LLM responds based on the given context or document. This means that all responses should be consistent with and grounded within that document, even when the model's internal knowledge might conflict with it. Faithfulness assesses how well an LLM's output adheres to the reasoning process during prediction or generation. In this sense, faithfulness demands that responses are grounded in the input data or externally retrieved documents, ensuring that the model does not hallucinate additional information. It is especially vital in applications such as RAG systems, where LLMs integrate retrieved documents into their response generation processes (Lewis et al., 2020). Research on LLM unfaithfulness has explored various challenges, including maintaining faithfulness in abstractive summarization (Maynez et al., 2020) and generating truthful responses in question-answering systems (Lin et al., 2022). Studies have investigated how manipulating chain-of-thought prompts can reveal adherence to intended reasoning processes (Lanham et al., 2023; Qiu et al., 2024), while others have examined how introducing biasing features in prompts can lead to unfaithful

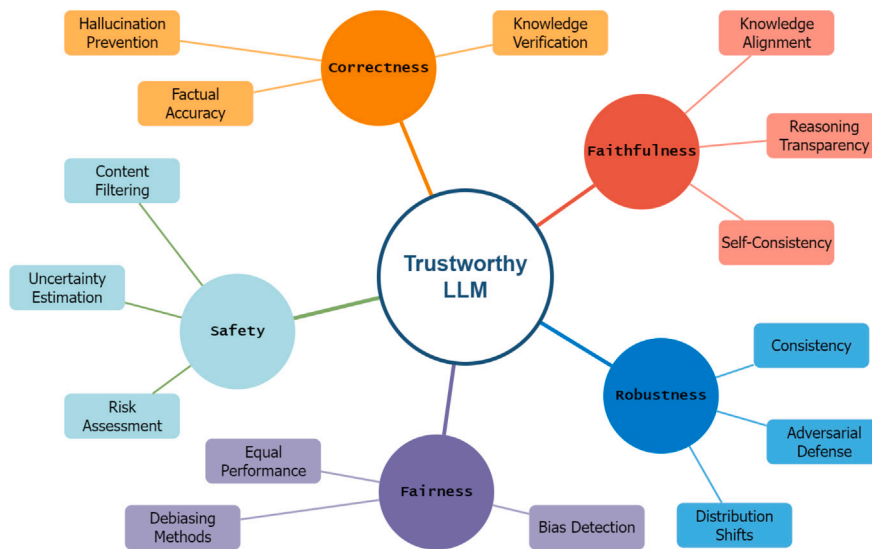


Fig. 6. Hierarchical breakdown of the LLMs trustworthiness framework, illustrating the main components and their subcomponents.

explanations (Turpin et al., 2023). Despite significant progress in understanding the mechanisms behind unfaithfulness, the challenge remains to develop robust models that consistently produce faithful and explainable responses. Addressing this challenge is vital for maintaining user trust and mitigating the spread of misinformation (Xylogiannopoulos et al., 2024).

Recent research suggests that more capable LLMs such as GPT-4 and Llama2-70b tend to demonstrate higher trustworthiness. These models perform well in tasks that require moral judgment and are more adept at defending against adversarial attacks. However, they also encounter challenges such as over-alignment, where they may become excessively cautious in their responses. This highlights the need for improved training methods that emphasize contextual understanding over rigid adherence to rules. Proprietary models like ChatGPT and GPT-4 generally exhibit higher trustworthiness compared to their open-source counterparts. Nevertheless, successful open-source models like Llama2 showcase the potential for significant contributions to the development of trustworthy AI, emphasizing that advancements in trustworthiness are possible across different platforms.

6.2. Limitations and directions

While significant progress has been made in red teaming LLMs, the current state of research has several limitations that need to be addressed. This subsection critically examines these limitations and outlines open research directions to advance the field.

Incomplete understanding of LLM vulnerabilities: Current research on LLM risks tends to be fragmented and lacks a comprehensive understanding of the full spectrum of vulnerabilities in LLMs. Most studies focus on specific aspects, such as bias or adversarial attacks, without a comprehensive view of all potential risks. Future research should aim to develop a holistic framework that categorizes and inter-relates various risks factors, facilitating a more thorough understanding of LLM vulnerabilities. This limitation also hampers the development of holistic LLM security frameworks that can address multiple vulnerabilities simultaneously, under a generalized framework. Hence, exploring the interaction between different types of risks and their cumulative impact on LLM performance and trustworthiness is a critical area that remains underexplored. Additionally, while our Section 4 briefly highlights some fundamental theoretical explanations linking vulnerabilities to inherent LLM weaknesses; such as statistical associations, contextual manipulation susceptibility, and challenges in intent inference. We acknowledge that a comprehensive theoretical exploration is still necessary.

To address these limitations, we suggest future research to prioritize the development of systematic frameworks to analyze the root causes of attacks and link them to LLM vulnerabilities beyond purely descriptive attack taxonomies. Such frameworks may formalize LLM weaknesses based on fundamental LLM design and architectural limitations, including reliance on statistical correlations, challenges in accurately recognizing user intent, and susceptibility to contextual manipulation. Employing causal modeling can further clarify how these inherent weaknesses lead to specific attack vectors. Moreover, building a theoretical foundation that links LLMs design decisions to security risks would enable proactive vulnerability assessment. Finally, integrating dynamic attacks mitigation or blue teaming strategies can shift LLM security from reactive countermeasures toward proactive and adaptive defenses.

Limited scope of red teaming attacks: The majority of red teaming techniques focus on a limited set of attack vectors, primarily prompt-based and data manipulation attacks. While these are important, they do not encompass the full spectrum of potential threats. For example, model exploitation attacks and information extraction attacks are less studied but pose significant risks. Additionally, the techniques are often evaluated in controlled settings that may not fully capture real-world complexities. Expanding the scope of red teaming to include these and other emerging attack vectors is essential. Furthermore, developing more sophisticated attack scenarios that mimic real-world threats can provide deeper insights into LLM vulnerabilities.

Furthermore, While our proposed taxonomy of LLMs red teaming attacks offers a structured framework for understanding and representing various adversarial strategies, it is not without limitations. As the domain is consistently evolving with new attack methods continually emerging, the proposed taxonomy based on current knowledge may not encompass future adversarial techniques, necessitating regular updates to remain comprehensive. In some cases, some attack strategies may fit into multiple categories within the taxonomy, based on the dynamic attack vector. The effectiveness and categorization of certain attacks can vary depending on the specific application or deployment context of the LLM. These contextual nuances might be crucial for attacks classification under the taxonomy. The absence of standardized practices in LLMs red teaming complicates the consistent application of taxonomies for red teaming attacks classification. Our taxonomy primarily addresses currently identified vulnerabilities and attack vectors. This focus may not properly accommodate novel or unconventional threats that do not fit neatly into existing categories, which again underscores the call for continuous research and adaptation.

Gaps in evaluation methods and benchmarks: Evaluation methods and benchmarks for red teaming LLMs are still evolving. Existing metrics often fail to capture the multifaceted nature of LLM performance, especially in adversarial contexts. The evaluation of red teaming efforts and mitigation strategies is often inconsistent due to the lack of standardized benchmarks and metrics. Different studies use varied datasets, evaluation criteria, and performance measures, making it difficult to compare results and draw generalized conclusions. This lack of standardization undermines the reliability of research findings and impedes progress in developing robust evaluation frameworks. Future work should aim to develop comprehensive evaluation frameworks that incorporate diverse metrics, including ethical and societal impacts. Establishing standardized benchmarks for various attack and mitigation scenarios will also enhance the comparability and reproducibility of research findings.

Exclusion of red teaming defenses and mitigation strategies: The focus of this review is primarily on the identification and analysis of exploits in LLMs through red teaming methodologies. While this approach is crucial for uncovering potential weaknesses, it inherently emphasizes the offensive aspects of security assessment. Consequently, we do not delve into defense or mitigation strategies against these identified exploits. The landscape of defense mechanisms for LLMs is vast and continually evolving, addressing a multitude of attack vectors and vulnerabilities. Each red teaming attack may exploit different weaknesses within an LLM which necessitates tailored defense strategies. There is not always a straightforward one-to-one correspondence between specific attacks and their countermeasures. Instead, a single defense mechanism might mitigate multiple vulnerabilities, or conversely a particular vulnerability might require a combination of defenses. This complexity renders the comprehensive coverage of defense strategies beyond the scope of our current work. Given the dynamic nature of defense methodologies, future research should aim to systematically develop and evaluate these strategies. Developing a comprehensive framework that maps specific vulnerabilities to corresponding defenses might be invaluable. Such a framework could guide practitioners in selecting appropriate mitigation techniques tailored to the unique threat landscape of their LLM applications.

Emerging trends and global challenges in LLM red teaming: Our review identifies several emerging trends that call for further analysis. Recent advances such as cross-modal attacks and stealth backdoors are beginning to reshape the threat landscape for LLMs. Cross-modal attacks, which exploit vulnerabilities by combining multiple input modalities (such as text, images, and audio) challenge traditional evaluation frameworks that are primarily designed for single-modality attacks. These attacks reveal complex interactions between different data types, raising new concerns about the integrity and alignment of multi-modal systems. Similarly, stealth backdoors engineered to remain undetected during regular evaluations present unique risks by embedding malicious behaviors that only manifest under specific conditions. Beyond these technical developments, there is a notable gap in addressing global challenges. Most current studies focus on high-resource languages and mainstream cultural contexts, leaving the vulnerabilities of LLMs in low-resource languages or culturally diverse settings underexplored. This oversight can hinder the applicability of existing red teaming methods in real-world deployments across diverse environments. Furthermore, while our review extensively discusses emerging red teaming methodologies from an academic perspective, we acknowledge that practical validation through real-world case studies or documented examples of successful red teaming implementations remains limited. Integrating such case studies would enhance the practical relevance of the research by illustrating how these advanced attack methodologies perform in operational settings and by highlighting the real-world challenges of scalability, generalizability, and adaptability. Bridging these gaps requires not only the continuous evolution of academic research but also stronger connections with industry practices to develop robust evaluation frameworks for real-world systems.

Insufficient focus on long-term safety and ethical implications: The long-term safety and ethical implications of deploying LLMs are areas that require more attention. Current research tends to focus on immediate technical challenges, often overlooking the broader societal impacts. Issues such as the potential for misuse, unintended consequences, and ethical dilemmas need to be addressed in a more structured manner. Future research should integrate ethical considerations into the core of LLM development and deployment processes. This includes developing frameworks for responsible AI governance, establishing ethical guidelines, and conducting longitudinal studies to understand the long-term effects of LLM deployment. This includes development of frameworks that ensure fairness, transparency, and accountability. Additionally, the development of LLMs faithfulness discretion needs to be prioritized, ensuring that generated responses accurately reflect the input and context.

Need for interdisciplinary collaboration: The complexity of LLM risks and the diversity of potential attack vectors necessitate interdisciplinary collaboration. Current research efforts are often siloed within specific domains, limiting the breadth of insights and solutions. Encouraging collaboration between experts in computational linguistics, machine learning, cybersecurity, ethics, and social sciences can lead to more comprehensive and innovative approaches to red teaming LLMs. Future research should encourage interdisciplinary projects and platforms for knowledge exchange among different fields.

By addressing these limitations and pursuing the outlined open directions, the research community can significantly enhance the robustness, security, and ethical deployment of safe LLMs. This will ensure that LLMs continue to evolve as reliable and trustworthy tools in various applications.

7. Implications of this study

We have thoroughly examined the landscape of red teaming for LLMs in this comprehensive review; focusing specifically on vulnerability exploits, potential attacks, risk evaluation methods, and benchmarks. Our exploration, guided by well-defined research questions, has provided critical insights expected to advance both current knowledge and the literature on LLM security.

In this section, we highlight the theoretical and practical implications of our study, drawing on key insights from our comprehensive analysis of LLM red teaming attacks and evaluation methods. We show how our findings deepen the understanding of LLM vulnerabilities and identify opportunities to expand and strengthen red teaming research. This includes building on the structured taxonomy we introduced and our review of risk assessment frameworks. Our aim is to offer a foundational perspective to guide both academic research and practical efforts in securing LLMs.

The following subsections synthesize key findings from our review, outline future opportunities, and reflect on the broader impact of our work.

7.1. Synthesis of key findings and insights

A central outcome of this review is the introduction of a structured and intuitive taxonomy for red teaming attacks on LLMs. This taxonomy organizes attack types of prompt-based manipulation, data poisoning, model exploitation, information extraction, and performance degradation into a coherent framework. By doing so, it clarifies the threat landscape and supports both the design of targeted red teaming strategies and the development of defenses tailored to specific vulnerabilities.

Alongside this, our review highlights the fragmented state of current red teaming research. Many studies focus on isolated risk dimensions; such as adversarial prompts, safety alignment, or bias, without offering a holistic perspective on the full range of vulnerabilities. This fragmentation impedes the development of unified security frameworks capable of addressing multiple attack surfaces, highlighting the need for integrated threat models and cross-cutting defensive strategies.

Moreover, while many works describe specific vulnerabilities and attack techniques, few explicitly connect them to the deeper theoretical limitations of LLMs, such as statistical pattern dependence, context sensitivity, or limited capacity for intent understanding. Our taxonomy thus not only structures existing knowledge but also reveals a critical gap: the need for future research to bridge practical exploits with the architectural constraints that give rise to them.

We also conducted a critical analysis of evaluation methodologies used in red teaming studies. Current practices suffer from inconsistency, with heterogeneous use of datasets, benchmarks, and metrics. This lack of standardization hinders meaningful comparisons across studies and impedes cumulative scientific progress. We emphasize the importance of developing scalable, standardized, and multidimensional evaluation frameworks that can reliably assess LLM vulnerabilities in varied contexts.

Our review further indicates a pervasive linguistic and cultural bias in LLM red teaming research. Most efforts rely on English-language datasets and evaluation settings, with minimal attention paid to low-resource languages or cross-cultural contexts. This skews the understanding of LLM vulnerabilities and undermines the generalizability of red teaming findings as these systems are deployed globally.

Relatedly, many benchmark datasets used for red teaming and evaluation inherit biases from their sources, often based on web or social media content. These biases can reinforce harmful model behaviors while concealing underlying vulnerabilities. Improving the quality and representativeness of datasets is therefore essential for achieving fair and accurate assessments of LLM robustness.

Finally, we identify a notable disconnect between academic research and real-world industry practice. While advances in red teaming techniques and mitigation strategies continue to emerge in academic settings, their validation in operational environments remains limited. Bridging this gap is crucial to ensure that research-driven innovations translate into practical improvements in the safety, reliability, and trustworthiness of deployed LLMs.

7.2. Opportunities for advancing LLM red teaming

This subsection outlines emerging opportunities to strengthen and expand the field, addressing limitations identified in the current landscape of LLM red teaming research. One important direction is the development of comprehensive theoretical frameworks that systematically link observed vulnerabilities to underlying model properties, by formalizing the connection between red teaming exploits and foundational weaknesses such as reliance on statistical correlations, susceptibility to contextual manipulation, and challenges in intent inference. Establishing such strong theoretical basis should enable the design of more principled attack methodologies and more robust defense mechanisms.

Expanding the scope of red teaming activities represents another important opportunity. With LLMs increasingly integrating multimodal inputs; emerging attack vectors such as cross-modal adversarial examples and stealth backdoors demand greater attention. Future research should focus on developing red teaming strategies that address vulnerabilities across text, images, audio, and other modalities to ensure that security assessments contend with technological advances. In parallel, expanding research into low-resource languages and diverse cultural background settings is key for globally resilient AI ecosystem.

Standardization of evaluation methods remains an urgent challenge for advancing the comparability and reproducibility of red teaming research. The domain would benefit greatly from unified benchmarking protocols, shared datasets, and consensus metrics that rigorously evaluate model vulnerabilities across a variety of attack types and operational contexts. Such standardization would improve scientific rigor and accelerate cumulative knowledge building across research groups and institutions in academia and industry.

Beyond identification of vulnerabilities, a vital step involves integrating defensive strategies more tightly with red teaming efforts. Developing structured framework(s) that map specific attack vectors to corresponding defense techniques would strengthen the practical utility of red teaming exercises, and would enable researchers and practitioners to systematically test and refine mitigation approaches jointly with attack simulations.

The opportunities also exist in improving the quality and diversity of datasets used for LLM red teaming. Potentially, future datasets should prioritize balance across languages, domains, and cultural contexts; while actively mitigating biases originating from their reliance on web-scraped or social media content. Building such datasets may elevate the fairness, reliability, and relevance of red teaming evaluations for more trustworthy LLM deployments.

Another key area for future work is in addressing long-term safety and ethical implications of LLMs. Future research should consider the broader societal impacts of AI deployment beyond immediate technical challenges; including responsible governance frameworks, ethical guidelines for red teaming practices, and longitudinal studies tracking the evolving risks associated with increasingly capable models.

Finally, interdisciplinary collaboration is essential for attending the complex and multifaceted LLM-oriented risks. Bridging expertise across machine learning, cybersecurity, computational linguistics, ethics, and social sciences will facilitate the design of more holistic red and blue teaming strategies that anticipate a wider range of adversarial behaviors and societal concerns. Particularly, interdisciplinary red teaming exercises that incorporate human factors, regulatory insights, and system-level considerations may enhance the robustness and trustworthiness of advancing LLM systems.

7.3. Impact statement

This study contributes to the evolving understanding of red teaming techniques for LLMs by surveying current approaches, identifying key evaluation challenges, and presenting a structured framework to guide future research and practice. This work addresses critical gaps in the assessment of LLM security and robustness by providing a comprehensive taxonomy of attack vectors, along with a review of systematic evaluation methods and benchmarks.

One of our framework's key strengths is its structured and comparative analysis of red teaming strategies. This systematic taxonomy not only facilitates more rigorous testing of model vulnerabilities but also supports clearer communication of security issues across academic, industrial, and regulatory communities. Another important aspect is that the framework is adaptable, making it particularly valuable in rapidly evolving threat landscapes where new adversarial techniques continuously emerge.

While the proposed taxonomy and evaluation strategies offer both theoretical and practical contributions, several areas warrant further validation, particularly regarding the scalability and generalizability of automated evaluation pipelines across diverse LLM architectures and deployment contexts. Although our study provides extensive benchmarks and evaluation metrics, practical environments may involve complex, context-specific vulnerabilities not fully captured by current datasets or automated tools. Additional evaluation insights and case studies may further enhance the framework's applicability and validity in operational settings. Furthermore, the practical implementation of the proposed red teaming framework in industry or policy contexts must consider resource constraints, such as computational cost and expertise requirements. Future research should explore ways to streamline evaluations and incorporate user-friendly interfaces or guidelines, making the framework accessible beyond academia to practitioners and policymakers.

In a broader context, our contributions support the development of safer, more resilient LLMs by providing a foundational basis for evaluating and mitigating potential harms, and guiding future research. As LLMs become increasingly integrated into sensitive applications, the implications of our work extend significantly beyond academia, informing policies and practices around responsible AI deployment. Ultimately, our work emphasizes the continuous need for vigilance in LLM security; highlighting the importance of continued research, validation, and collaboration to ensure LLMs are trustworthy, secure, and aligned with societal values.

8. Conclusion

LLMs are increasingly pivotal across various domains. However, despite their transformative capabilities, these models present significant vulnerabilities that expose them to various forms of attack. Understanding these vulnerabilities and developing robust safeguards is essential for ensuring their secure and responsible deployment. This review offers a comprehensive and novel examination of LLM red teaming, distinguished by its broad scope and intuitive organization. We systematically explore a range of red teaming attacks and arrange them in a novel and intuitive taxonomy; including prompt-based attacks, data manipulation attacks, model exploitation attacks, information extraction attacks, and model degradation attacks. Additionally, we explore evaluation

methods and benchmarks, assessing datasets, evaluation metrics, and benchmarking methods used in LLM red teaming and risk assessment. A significant contribution of our review is the integration of recent and impactful research. We ensure that our review reflects the current state of LLM security and provides new insights alongside established methods by incorporating cutting-edge studies and developments. The structured presentation of our findings offers a comprehensive and actionable resource. Our clear organization of red teaming methods and evaluation frameworks, and their insights, facilitates a deeper understanding of the involved complexities and provides practical guidance for future research and applications. This review underscores the importance of addressing these challenges and serves as a valuable guide for researchers, practitioners, and policymakers. By addressing identified gaps and applying the proposed frameworks, we may advance toward more secure and responsible AI systems.

CRedit authorship contribution statement

Muhammad Shahid Jabbar: Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Sadam Al-Azani:** Conceptualization, Data curation, Formal analysis, Methodology, Supervision, Validation, Writing – review & editing. **Abrar Alotaibi:** Data curation, Formal analysis, Investigation, Writing – original draft. **Moataz Ahmed:** Conceptualization, Formal analysis, Funding acquisition, Project administration, Supervision, Validation, Writing – review & editing.

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Data availability

No data was used for the research described in the article.

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